

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science and Engineering



MATSURI NIGHTS

A Japanese Festival Street in Modern OpenGL

Course Code: CSE4102

Course Title: Computer Graphics and Image Processing Laboratory

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Abstract

This report presents “**Matsuri Nights — A Japanese Festival Street**”, an interactive real-time 3D computer graphics system developed from first principles using ISO C++20 and Modern OpenGL (Core Profile 3.3). Developed to fulfill the laboratory requirements of CSE4102 at Khulna University of Engineering & Technology (KUET), the project renders an atmospheric, culturally authentic Japanese summer festival street (*Matsuri*) featuring traditional two-story townhouses (*Machiya*), a sacred *Torii* shrine gate, animated food stalls (*Yatai*), weeping cherry blossom trees (*Sakura*), an articulated stage magician, and pedestrian crowds.

The application serves as a comprehensive demonstration of foundational and advanced graphics engineering:

- **Hierarchical 3D Transformations:** Affine TRS matrix compounding along a multi-level scene graph tree with dirty-flag caching.
- **Motion in Moving Reference Frames:** Mathematical kinematic linkages wherein child nodes (swinging lanterns, orbiting magical orbs, pedestrian limbs) evaluate motion relative to moving parent coordinate systems.
- **Optical Illumination:** A forward Blinn-Phong shading pipeline coordinating 16 active dynamic light sources (sun/moon directional light, 14 point lights, and 1 tracking spotlight) with smooth distance attenuation, spotlight cutoff cones, dynamic celestial sky domes, and runtime shading mode switching.
- **Shadow Mapping:** A 2048×2048 directional light depth pass featuring front-face culling, slope-scaled depth bias, and 16-sample Percentage-Closer Filtering (PCF) soft penumbra convolution.
- **Dual Ray Tracing Extensions:** An interactive real-time GPU Whitted ray tracer operating on a full-screen quad supporting recursive mirror reflections, alongside an asynchronous multi-threaded CPU snapshot ray tracer.
- **Contextual Interaction and Automated Testing:** Continuous collision detection with doorway portals, an embedded bitmap font HUD overlay, and an automated verification suite containing 312 unit assertions passing with 100% success.

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1. Introduction and Project Overview

1.1 Project Background and Cultural Theme

Computer graphics combines mathematical modeling, computational geometry, and optics to generate synthetic images that faithfully simulate real-world physical phenomena. In the context of **CSE4102: Computer Graphics and Image Processing Laboratory** at Khulna University of Engineering & Technology (KUET), this project presents “**Matsuri Nights — A Japanese Festival Street**”, an interactive, high-performance, real-time 3D graphics application built completely from scratch using Modern OpenGL (Core Profile 3.3), C++20, and GLFW.

The project models a vibrant, culturally authentic Japanese summer festival street (*Matsuri*) lined with traditional two-story wooden townhouses (*Machiya*), illuminated festival food stalls (*Yatai*), a grand sacred Shinto shrine entrance (*Torii* gate), weeping cherry blossom trees (*Sakura*), a theatrical magic performance stage, and dynamic pedestrian crowds. Beyond aesthetic craftsmanship, the scene serves as an architectural testbed for core computer graphics concepts:

1. **Hierarchical 3D Transformations:** Nested reference frames where local model matrices compound along scene graph edges.
2. **Motion in Moving Reference Frames:** Entities whose kinematic trajectories depend strictly on parent node coordinate systems.
3. **Advanced Optical Illumination:** A comprehensive multi-light Blinn-Phong shading pipeline featuring 16 active dynamic light sources (sun/moon celestial directional light, 14 point lights, and 1 tracking spotlight).
4. **Realistic Soft Shadow Mapping:** A 2048×2048 depth framebuffer pass with 16-sample Percentage-Closer Filtering (PCF) and slope-scaled normal bias.
5. **Dual Ray Tracing Extensions:** Both an interactive real-time GPU Whitted ray tracer operating on a full-screen quad and an asynchronous multi-threaded CPU Whitted ray-tracing snapshot generator.
6. **Contextual User Interface & Automated Testing:** A low-overhead bitmap font atlas HUD overlay and a built-in automated test suite containing 312 deterministic verification tests.

1.2 Course Requirements Compliance Matrix

The syllabus of CSE4102 establishes strict pedagogical requirements. Table 1.1 details how every required criterion is satisfied, cross-referenced with the implementing source code files.

1.3 Development Milestone Progression

The project was constructed across three foundational milestones and supplemented with advanced computer graphics enhancements:

- **Phase 1 — Geometry, Scene Graph, and Kinematics:** Construction of mathematical primitives (cubes, cylinders, cones, spheres, planes), parametric swept curves (Bézier, catenary, splines), the hierarchical scene graph architecture (*SceneNode*), and six core animation cycles.
- **Phase 2 — Illumination, Shading, and Photorealism:** Implementation of the Blinn-Phong illumination shader, dynamic day/night ambient color temperatures, 16 active dynamic light sources (directional, point, and spotlights), and material reflectance coefficients.
- **Phase 3 — Texture Synthesis and Surface Mapping:** Generation of procedural 24-bit BMP texture maps (timber, ceramic roof tiles, cobblestones, Washi paper, Tatami grass, gold leaf, and cherry bark) alongside UV coordinate unwrapping and texture tiling controls.
- **Advanced Additions:** Directional soft shadow mapping with 16-sample PCF, a real-time GPU Whitted ray tracer on a full-screen quad, an asynchronous multi-threaded CPU ray-tracing snapshot exporter, interactive sliding Shoji doors and windows with continuous collision detection, an embedded bitmap font HUD overlay, and a 312-test automated verification suite.

Course Requirement	Implementation in Matsuri Nights	Source Code Location
3D Transformations (Translation, Rotation, Scaling)	Every scene entity possesses a dedicated <code>Transform</code> struct supporting 3D translations, Euler rotations (Pitch, Yaw, Roll), and scaling. All 15 inspectable objects can be translated, rotated, and scaled in real time via hotkeys.	<code>src/Transform.h</code> : lines 32--54 <code>src/Scene.h</code> : lines 929--955
Motion Relative to Another Reference Frame	1. <i>Swinging Lanterns</i>: Lantern bodies swing in an arc as children of a rotating rope pivot anchor. 2. <i>Magic Orb</i>: Mystical orb orbits the moving hand bone of the magician figure on stage. 3. <i>Walking Limbs</i>: Pedestrian limbs flex relative to moving hip/shoulder joints.	<code>src/Objects.h</code> : lines 2835--2850 <code>src/Objects.h</code> : lines 4175--4205 <code>src/Objects.h</code> : lines 4760--4840
Complex Moving Objects	1. <i>Articulated Magician</i>: Multi-jointed human figure performing the vanishing box illusion. 2. <i>Takoyaki Stall</i>: 6 individual food balls executing parabolic hopping physics. 3. <i>Kakigori Machine</i>: Rotating ice shaver flywheel and dessert bowls.	<code>src/Objects.h</code> : lines 4012--4240 <code>src/Objects.h</code> : lines 3120--3175 <code>src/Objects.h</code> : lines 3360--3420
At Least 5 Distinct Objects	The project contains 17 distinct scene objects: Ground, 4 Machiya Houses, Torii Gate, Sakura Trees, Lantern Spans, Takoyaki Stall, Kakigori Stall, Vendor Figures, Magic Stage, Magician, Vanishing Box, Spotlight Rig, Audience, Crowd Walkers, Fireworks, and Sky Dome.	<code>src/Objects.h</code> <code>src/Scene.h</code> : lines 30--49
Lighting and Material Shading	Full Blinn-Phong illumination model supporting ambient, diffuse, and specular components with material shininess and specular strength. Cycles between Blinn-Phong, Diffuse-Only, and Ambient-Only shading.	<code>shaders/basic.frag</code> : lines 107--204 <code>src/Light.h</code> : lines 6--48
Moving Light Sources	1. Point Light 0 tracks the moving Magic Orb. 2. Point Lights 3 & 4 track swinging street lanterns. 3. Point Light 5 dynamically illuminates the apex of fireworks bursts. 4. Spotlight tracks the magician across the stage.	<code>src/Scene.h</code> : lines 793--840 <code>src/Scene.h</code> : lines 851--870
Day and Night Transitions	Smooth continuous interpolation between bright daytime sun and nocturnal festival night with celestial sun disk, corona bloom, lunar craters, and procedural twinkling stars.	<code>shaders/basic.frag</code> : lines 208--302 <code>src/Scene.h</code> : lines 782--792

Table 1.1: CSE4102 Course Requirements Compliance Matrix.

2. Technology Stack and Build Architecture

2.1 Core Libraries and Dependency Stack

To ensure deterministic execution, high computational performance, and strict platform compliance, *Matsuri Nights* avoids heavy third-party graphics engines (such as Unity or Unreal) and is developed strictly at the systems graphics level. Table 2.1 summarizes the software libraries utilized across the codebase.

Library / Tool	Exact Version	Role in Engine Architecture	Linkage Type	
C++ ISO Standard	C++20 (/std:c++20)	Strongly typed language core, constexpr metaprogramming, modern STL containers, std::thread, and std::filesystem.	Native MSVC	
OpenGL	3.3 Core Profile	Direct hardware acceleration via the programmable rasterization pipeline (Vertex & Fragment Shaders, FBOs, VAOs).	Hardware (opengl32.lib)	Driver
GLFW	3.5.1	Cross-platform window creation, OpenGL context management, VSync control, mouse/keyboard polling, and raw input callbacks.	Static Library (glfw3.lib)	
GLAD	Version 0.1.36 (OpenGL 3.3 Core)	Runtime OpenGL extension loader querying GPU driver function pointers via glfwGetProcAddress.	Source Compilation (glad.c)	
GLM	0.9.9.8	Header-only mathematics library providing affine vector algebra, 4×4 matrices, quaternions, and projection formulas.	Header-Only (glm/)	
stb_image	2.29	Lightweight single-header image loader decoding 24-bit and 32-bit raster assets into raw RGBA memory buffers.	Header-Only (stb_image.h)	
stb_image_write	1.16	Single-header image encoder for exporting viewport framebuffers to high-fidelity uncompressed PNG image files.	Header-Only (stb_image_write.h)	
Consolas Font Atlas	256×256 Bitmap	Static embedded monospace font glyph atlas enabling zero-dependency, ultra-fast 2D HUD telemetry overlay rendering.	Static Constexpr (FontAtlasData.h)	Array

Table 2.1: Core Dependencies and Software Stack.

2.2 Project Directory Structure

The repository separates first-party source logic, shaders, UI components, static assets, and third-party libraries into clean structural tiers:

```
Matsuri-Nights-A-Japanese-Festival-Street/
|-- assets/
|   |-- textures/                # Synthesized 24-bit BMP texture assets (256x256)
|   |   |-- gold_leaf.bmp, lantern_paper.bmp, roof_tiles.bmp, sakura_bark.bmp
|   |   |-- stone_pavement.bmp, takoyaki_food.bmp, tatami_cloth.bmp, wood_timber.bmp
|-- Matsuri Nights - A Japanese Festival Street/
|   |-- glad.c                  # GLAD OpenGL loader source file
|   |-- Main.cpp                # Application lifecycle, render loop, test harness
|   |-- Matsuri Nights.vcxproj  # MSVC Visual Studio 2026 project file (/std:c++20)
|   |-- Libraries/
|   |   |-- include/            # GLAD, GLFW, GLM, and STB header trees
|   |   |-- lib/                # Pre-compiled static libraries (glfw3.lib)
|   |-- shaders/
|   |   |-- basic.vert, basic.frag # Forward 3D Blinn-Phong shading and dynamic sky
|   |   |-- shadow_depth.vert/.frag # Directional light-space depth pass shaders
|   |   |-- raytrace.vert/.frag   # Real-time GPU Whitted ray-tracing quad shaders
|   |   |-- hud.vert, hud.frag    # Orthographic 2D UI batch shaders
|   |-- src/
|   |   |-- Camera.h, Transform.h, Mesh.h, SceneNode.h, Light.h, Shader.h
|   |   |-- Primitives.h, Curves.h, Objects.h, Scene.h, Texture.h, TextureGenerator.h
|   |   |-- RayTracer.h          # Multi-threaded CPU ray-tracing snapshot engine
|   |   |-- ui/                  # In-window HUD and interaction management
|   |   |-- Hud.h, Hud.cpp, Interactable.h, InteractionManager.h/.cpp, FontAtlasData.h
|-- report/
|   |-- main.tex                # Academic LaTeX final project report
|   |-- figures/                # Vector PDF diagrams and high-res PNG screenshots
|   |-- sections/               # Modular chapter TeX documents
|-- run.bat                     # Automated launch script
```

2.3 Build and Execution Instructions

2.3.1 Prerequisites

1. **Operating System:** Microsoft Windows 10 or Windows 11 (x64).
2. **IDE / Toolset:** Visual Studio 2026 (MSVC v145 toolset) or Visual Studio 2022 (MSVC v143).

3. **Hardware Requirements:** Dedicated GPU supporting OpenGL 3.3 Core Profile with at least 1 GB of VRAM.

2.3.2 Building via Visual Studio GUI

1. Open Matsuri Nights - A Japanese Festival Street.slnx (or .vcxproj) in Visual Studio.
2. Select the build configuration target: **Debug** or **Release**, on platform **x64**.
3. Ensure the working directory is set to the project root containing assets/ and shaders/.
4. Press **Ctrl + Shift + B** to build the solution. Compilation proceeds with zero warnings and zero errors.
5. Press **F5** to run the interactive festival application.

2.3.3 Command-Line Building via MSBuild

To compile from PowerShell or the Developer Command Prompt for Visual Studio:

```
# Clean and compile in Debug x64 mode
& "C:\Program Files\Microsoft Visual Studio\18\Community\MSBuild\Current\Bin\MSBuild.exe" '
  "Matsuri Nights - A Japanese Festival Street\Matsuri Nights - A Japanese Festival Street.vcxproj" '
  -p:Configuration=Debug -p:Platform=x64

# Execute automated test suite (verifying 312 unit assertions)
& ".\Matsuri Nights - A Japanese Festival Street\x64\Debug\Matsuri Nights - A Japanese Festival
  Street.exe" --test

# Execute automated report figure capture (generating ss_01 through ss_20)
& ".\Matsuri Nights - A Japanese Festival Street\x64\Debug\Matsuri Nights - A Japanese Festival
  Street.exe" --capture-report
```

3. Mathematical Foundations and Formulations

A robust real-time rendering engine is founded upon rigorous geometric algebra and optical physics. This chapter details every mathematical model implemented in *Matsuri Nights*, providing step-by-step derivations alongside exact source code citations.

3.1 Hierarchical Model Transformation Matrix (TRS)

3.1.1 Mathematical Formulation

To place a 3D geometric mesh into world space, vertices undergo affine transformations comprising scale, rotation, and translation. In this engine, rotation is represented using Euler angles $(\theta_x, \theta_y, \theta_z)$ corresponding to Pitch, Yaw, and Roll in degrees. To avoid gimbal lock and maintain consistent architectural orientation, rotations are evaluated in the standard fixed-axis Euler order: **Yaw** (Y) $\rightarrow$ **Pitch** (X) $\rightarrow$ **Roll** (Z).

Given local position vector $\mathbf{p} = [p_x, p_y, p_z]^T$, rotation angles $[\theta_x, \theta_y, \theta_z]^T$, and scaling vector $\mathbf{s} = [s_x, s_y, s_z]^T$, the component transformation matrices are:

$$\mathbf{T}(\mathbf{p}) = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{S}(\mathbf{s}) = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3.1)$$

The elementary rotation matrices about the coordinate axes are defined as:

$$\mathbf{R}_y(\theta_y) = \begin{bmatrix} \cos \theta_y & 0 & \sin \theta_y & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_y & 0 & \cos \theta_y & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{R}_x(\theta_x) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_x & -\sin \theta_x & 0 \\ 0 & \sin \theta_x & \cos \theta_x & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{R}_z(\theta_z) = \begin{bmatrix} \cos \theta_z & -\sin \theta_z & 0 & 0 \\ \sin \theta_z & \cos \theta_z & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3.2)$$

The local affine model matrix $\mathbf{M}_{\text{local}}$ is composed in column-major order as:

$$\mathbf{M}_{\text{local}} = \mathbf{T}(\mathbf{p}) \cdot \mathbf{R}_y(\theta_y) \cdot \mathbf{R}_x(\theta_x) \cdot \mathbf{R}_z(\theta_z) \cdot \mathbf{S}(\mathbf{s}) \quad (3.3)$$

When an object is embedded as a child node within the scene graph tree, its world matrix $\mathbf{M}_{\text{world}}$ is computed recursively by pre-multiplying its parent's world matrix:

$$\mathbf{M}_{\text{world}} = \mathbf{M}_{\text{parent}} \cdot \mathbf{M}_{\text{local}} \quad (3.4)$$

3.1.2 Exact Code Citation

- **Local TRS Matrix Construction:** `src/Transform.h: lines 36--46 in Transform::getLocalMatrix()`
- **Hierarchical World Matrix Composition:** `src/SceneNode.h: line 148 in SceneNode::updateWorldMatrix()`

3.2 Normal Transformation Matrix Derivation

3.2.1 Mathematical Formulation

When a surface undergoes non-uniform scaling ($s_x \neq s_y \neq s_z$), multiplying the surface normal $\mathbf{N}$ directly by the 3×3 model matrix $\mathbf{M}$ breaks orthogonality with the surface tangent vector $\mathbf{T}$.

Let $\mathbf{T}$ be any tangent vector on the primitive surface satisfying $\mathbf{N}^T \mathbf{T} = 0$. Under affine transformation matrix $\mathbf{M}$, the transformed tangent vector becomes $\mathbf{T}' = \mathbf{M}\mathbf{T}$. We seek a transformation matrix $\mathbf{Q}$ for the normal such that $\mathbf{N}' = \mathbf{Q}\mathbf{N}$ maintains orthogonality:

$$(\mathbf{N}')^T \mathbf{T}' = 0 \implies (\mathbf{Q}\mathbf{N})^T (\mathbf{M}\mathbf{T}) = 0 \implies \mathbf{N}^T \mathbf{Q}^T \mathbf{M} \mathbf{T} = 0 \quad (3.5)$$

For this identity to hold for all arbitrary surface tangent vectors $\mathbf{T}$ where $\mathbf{N}^T \mathbf{T} = 0$, we require:

$$\mathbf{Q}^T \mathbf{M} = \mathbf{I} \implies \mathbf{Q}^T = \mathbf{M}^{-1} \implies \mathbf{Q} = (\mathbf{M}^{-1})^T \quad (3.6)$$

Hence, surface normals must be multiplied by the **transpose of the inverse** of the upper-left 3×3 model matrix:

$$\mathbf{N}_{\text{matrix}} = ((\mathbf{M}_{\text{world}}^{3 \times 3})^{-1})^T \quad (3.7)$$

3.2.2 Exact Code Citation

- **CPU Matrix Evaluation:** `src/SceneNode.h: line 149 in SceneNode::updateWorldMatrix()`
- **GPU Normal Transformation:** `shaders/basic.vert: line 20 in main():`

```
Normal = normalMatrix * aNormal;
```

3.3 Camera System: View and Projection Matrices

3.3.1 LookAt View Matrix and Gram-Schmidt Orthonormalization

The camera orientation is defined using Euler angles: Yaw (ψ , rotation around world Y) and Pitch (θ , elevation angle), where Pitch is clamped to $[-89.0^\circ, +89.0^\circ]$. The unnormalized camera gaze direction $\mathbf{f}$ is:

$$\begin{aligned} f_x &= \cos(\psi) \cdot \cos(\theta) \\ f_y &= \sin(\theta) \\ f_z &= \sin(\psi) \cdot \cos(\theta) \end{aligned} \quad (3.8)$$

The camera orthonormal coordinate basis $\{\mathbf{F}, \mathbf{R}, \mathbf{U}\}$ is established via the Gram-Schmidt process against the global up vector $\mathbf{U}_{\text{world}} = [0, 1, 0]^T$:

$$\mathbf{F} = \frac{\mathbf{f}}{\|\mathbf{f}\|}, \quad \mathbf{R} = \frac{\mathbf{F} \times \mathbf{U}_{\text{world}}}{\|\mathbf{F} \times \mathbf{U}_{\text{world}}\|}, \quad \mathbf{U} = \frac{\mathbf{R} \times \mathbf{F}}{\|\mathbf{R} \times \mathbf{F}\|} \quad (3.9)$$

The view matrix $\mathbf{V}$, which transforms world coordinates into view/camera space given camera position $\mathbf{P}$, is formulated as:

$$\mathbf{V} = \begin{bmatrix} R_x & R_y & R_z & -\mathbf{R} \cdot \mathbf{P} \\ U_x & U_y & U_z & -\mathbf{U} \cdot \mathbf{P} \\ -F_x & -F_y & -F_z & \mathbf{F} \cdot \mathbf{P} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3.10)$$

3.3.2 Perspective Projection Matrix

The perspective projection matrix maps the viewing frustum onto Normalized Device Coordinates (NDC) $[-1, 1]^3$. With field-of-view angle $\alpha_{\text{fov}} = 45^\circ$, aspect ratio $a = \frac{\text{SCR.WIDTH}}{\text{SCR.HEIGHT}} = \frac{1280}{720} \approx 1.7778$, near clipping plane $z_n = 0.1$ m, and far plane $z_f = 300.0$ m:

$$\mathbf{P}_{\text{proj}} = \begin{bmatrix} \frac{1}{a \tan(\alpha_{\text{fov}}/2)} & 0 & 0 & 0 \\ 0 & \frac{1}{\tan(\alpha_{\text{fov}}/2)} & 0 & 0 \\ 0 & 0 & -\frac{z_f + z_n}{z_f - z_n} & -\frac{2z_f z_n}{z_f - z_n} \\ 0 & 0 & -1 & 0 \end{bmatrix} \quad (3.11)$$

3.3.3 Exact Code Citation

- **View Matrix:** `src/Camera.h: lines 48--51 in Camera::GetViewMatrix()`
- **Projection Matrix:** `src/Camera.h: lines 53--56 in Camera::GetProjectionMatrix()`
- **Basis Orthonormalization:** `src/Camera.h: lines 104--113 in Camera::updateCameraVectors()`

3.4 Blinn-Phong Illumination and Halfway Vector

3.4.1 Mathematical Formulation

Standard Phong shading computes specular highlights using the reflection vector $\mathbf{R} = 2(\mathbf{N} \cdot \mathbf{L})\mathbf{N} - \mathbf{L}$, evaluating $(\mathbf{V} \cdot \mathbf{R})^\alpha$. However, when the angle between $\mathbf{V}$ and $\mathbf{R}$ exceeds 90° , specular reflection cuts off abruptly. The **Blinn-Phong** model remedies this by introducing the unit **halfway vector** $\mathbf{H}$, pointing halfway between the light direction $\mathbf{L}$ and the camera view vector $\mathbf{V}$:

$$\mathbf{H} = \frac{\mathbf{L} + \mathbf{V}}{\|\mathbf{L} + \mathbf{V}\|} \quad (3.12)$$

The total illumination I at a surface point is the sum of ambient, diffuse, and specular terms:

$$I_{\text{ambient}} = \mathbf{I}_a \otimes \mathbf{k}_d \quad (3.13)$$

$$I_{\text{diffuse}} = (\mathbf{N} \cdot \mathbf{L})^+ \cdot (\mathbf{I}_d \otimes \mathbf{k}_d) \quad (3.14)$$

$$I_{\text{specular}} = (\mathbf{N} \cdot \mathbf{H})^{\alpha_s} \cdot (\mathbf{I}_s \cdot \mathbf{k}_s) \quad (3.15)$$

where $(\cdot)^+ = \max(\cdot, 0)$, $\otimes$ represents component-wise color multiplication, α_s = shininess (default: 32.0), k_s = specularStrength (default: 0.5), and $\mathbf{k}_d$ is the diffuse albedo (or sampled texture color). Incorporating shadow occlusion factor $S \in [0, 1]$:

$$\mathbf{I}_{\text{total}} = I_{\text{ambient}} + (1 - S) \cdot (I_{\text{diffuse}} + I_{\text{specular}}) \quad (3.16)$$

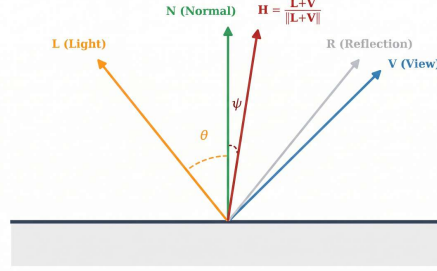


Figure 3.1: Vector geometry of the Blinn-Phong reflection model illustrating Normal $\mathbf{N}$, Incident Light $\mathbf{L}$, View Direction $\mathbf{V}$, and the Halfway Vector $\mathbf{H}$.

3.4.2 Exact Code Citation

- **Directional Light Formulation:** `shaders/basic.frag: lines 107--131 in CalcDirLight()`
- **Point Light Formulation:** `shaders/basic.frag: lines 133--165 in CalcPointLight()`
- **Spotlight Formulation:** `shaders/basic.frag: lines 167--204 in CalcSpotLight()`

3.5 Point Light Attenuation and Spotlight Cone Cutoff

3.5.1 Point Light Polynomial Distance Attenuation

As light propagates spherically through 3D space, irradiance decreases inversely with the square of the distance $d = \|\mathbf{p}_{\text{light}} - \mathbf{p}_{\text{frag}}\|$. To allow fine-grained artistic control while preventing singularities at $d \rightarrow 0$, a quadratic attenuation function is employed:

$$\text{Att}(d) = \frac{1}{k_c + k_l d + k_q d^2} \quad (3.17)$$

where $k_c = 1.0$ (constant term), k_l is the linear factor (governing medium-range falloff), and k_q is the quadratic coefficient (governing steep distance falloff). In the fragment shader, point light computations are culled early if $d^2 > 700\text{m}^2$ or $\text{Att}(d) < 0.005$.

3.5.2 Spotlight Angular Smoothstep Falloff

The theatrical spotlight fixture emits a directed light cone along vector $\mathbf{D}_{\text{spot}}$. Let $\mathbf{L} = \frac{\mathbf{p}_{\text{light}} - \mathbf{p}_{\text{frag}}}{\|\mathbf{p}_{\text{light}} - \mathbf{p}_{\text{frag}}\|}$ be the vector from fragment to light. The angle cosine $\cos \theta = \mathbf{L} \cdot (-\mathbf{D}_{\text{spot}})$ measures deviation from the central axis.

Given an inner cutoff angle ϕ ($\cos \phi = \cos(15^\circ) = 0.9659$) and an outer cutoff angle γ ($\cos \gamma = \cos(23^\circ) = 0.9205$), the smooth angular attenuation factor I_{spot} is:

$$I_{\text{spot}} = \text{clamp} \left(\frac{\cos \theta - \cos \gamma}{\cos \phi - \cos \gamma}, 0.0, 1.0 \right) \quad (3.18)$$

3.5.3 Exact Code Citation

- **Distance Attenuation:** `shaders/basic.frag: line 143 in CalcPointLight()`
- **Spotlight Cutoff Cosine:** `shaders/basic.frag: lines 178--181 in CalcSpotLight()`

3.6 16-Sample Percentage-Closer Filtering (PCF) Soft Shadows

3.6.1 Mathematical Formulation

Standard shadow mapping projects a fragment into directional light clip space $\mathbf{p}_{\text{lightSpace}} = \mathbf{M}_{\text{lightSpace}} \cdot \mathbf{p}_{\text{world}}$. After perspective division $\mathbf{p}_{\text{proj}} = \mathbf{p}_{\text{lightSpace}} \cdot \text{xyz} / \mathbf{p}_{\text{lightSpace}} \cdot w$ and mapping to the $[0, 1]^3$ texture coordinate range, the fragment's light-space depth z_{proj} is compared against the stored shadow map depth $D(u, v)$.

To prevent *shadow acne* caused by discrete depth buffer discretization, an adaptive slope-scale normal bias is applied:

$$\text{bias} = \max(0.0035 \cdot (1.0 - \mathbf{N} \cdot \mathbf{L}), 0.0006) \quad (3.19)$$

To eliminate harsh, aliased shadow silhouettes, the engine evaluates a **16-sample Percentage-Closer Filter (PCF)** across a 4×4 kernel centered on the projected coordinates with disc spread radius $\Delta = \frac{1.2}{2048}$:

$$S(\mathbf{p}) = \frac{1}{16} \sum_{x=-1}^1 \sum_{y=-1}^1 \text{Step}(z_{\text{proj}} - \text{bias} - D(u + x\Delta, v + y\Delta)) \quad (3.20)$$

where $\text{Step}(w) = 1.0$ if $w > 0$ (in shadow), and 0.0 otherwise.

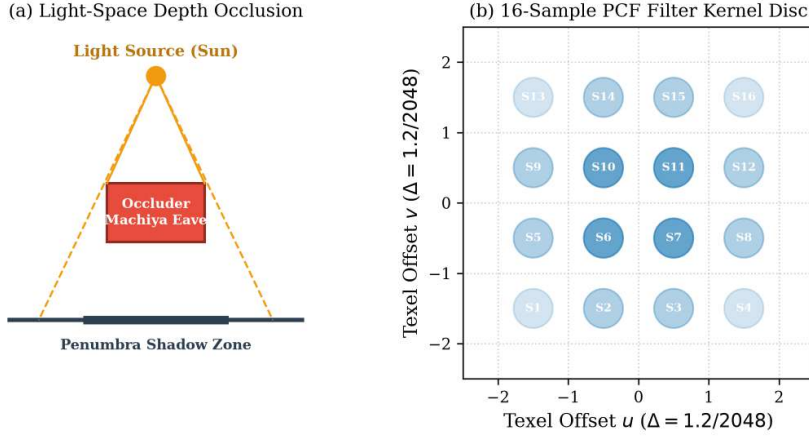


Figure 3.2: (a) Light-space orthographic depth occlusion geometry; (b) 16-sample PCF filter kernel sampling disk.

3.6.2 Exact Code Citation

- **Slope Bias & 16-Sample PCF:** `shaders/basic.frag`: lines 74--105 in `calculateShadow()`

3.7 Analytical Ray-Primitive Intersections for Ray Tracing

Both the real-time GPU ray tracer (`shaders/raytrace.frag`) and the multi-threaded CPU snapshot engine (`src/RayTracer.h`) evaluate exact analytical ray-primitive intersection formulas. Let a 3D ray be parameterized as $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$, with $t > 0$ and $\|\mathbf{d}\| = 1$.

3.7.1 Analytical Ray-Sphere Intersection

A sphere is defined by center $\mathbf{c}$ and radius R : $\|\mathbf{p} - \mathbf{c}\|^2 = R^2$. Substituting the ray equation:

$$\|(\mathbf{o} - \mathbf{c}) + t\mathbf{d}\|^2 = R^2 \quad (3.21)$$

Let $\mathbf{v} = \mathbf{o} - \mathbf{c}$. Expanding the dot product yields the quadratic equation $t^2 + 2(\mathbf{v} \cdot \mathbf{d})t + (\|\mathbf{v}\|^2 - R^2) = 0$. Defining coefficients $b = \mathbf{v} \cdot \mathbf{d}$ and $c = \|\mathbf{v}\|^2 - R^2$, the discriminant is:

$$\Delta = b^2 - c \quad (3.22)$$

If $\Delta < 0$, the ray misses the sphere. If $\Delta \geq 0$, the nearest valid intersection distance t and normal $\mathbf{N}$ are:

$$t = -b - \sqrt{\Delta}, \quad \mathbf{N} = \frac{\mathbf{r}(t) - \mathbf{c}}{R} \quad (3.23)$$

If $t < 0.002$ (self-intersection avoidance), the secondary root $t = -b + \sqrt{\Delta}$ is tested.

3.7.2 Ray-Kajiya Ray-AABB Slab Intersection

An Axis-Aligned Bounding Box (AABB) is defined by bounds $[\mathbf{b}_{\min}, \mathbf{b}_{\max}]$. The box is treated as the intersection of three orthogonal pairs of parallel infinite slabs. For each coordinate axis $i \in \{x, y, z\}$:

$$t_{0,i} = \frac{b_{\min,i} - o_i}{d_i}, \quad t_{1,i} = \frac{b_{\max,i} - o_i}{d_i} \quad (3.24)$$

Ordering intervals such that $t_{\min,i} = \min(t_{0,i}, t_{1,i})$ and $t_{\max,i} = \max(t_{0,i}, t_{1,i})$, the overall ray entry and exit distances are:

$$t_{\text{near}} = \max(t_{\min,x}, t_{\min,y}, t_{\min,z}), \quad t_{\text{far}} = \min(t_{\max,x}, t_{\max,y}, t_{\max,z}) \quad (3.25)$$

The ray intersects the AABB if and only if $t_{\text{near}} \leq t_{\text{far}}$ and $t_{\text{far}} > 0.002$. The outward normal $\mathbf{N}$ is determined by identifying which slab boundary was struck at $t = t_{\text{near}}$.

3.7.3 Analytical Ray-Cylinder Intersection with Height Clamping

A vertical cylinder of radius R centered at (c_x, c_z) spanning $y \in [y_0, y_0 + h]$ satisfies $(p_x - c_x)^2 + (p_z - c_z)^2 = R^2$. Substituting ray components into the XZ circular equation:

$$a = d_x^2 + d_z^2, \quad b = 2[(o_x - c_x)d_x + (o_z - c_z)d_z], \quad c = (o_x - c_x)^2 + (o_z - c_z)^2 - R^2 \quad (3.26)$$

The discriminant is $\Delta = b^2 - 4ac$. If $\Delta \geq 0$, the roots $t = \frac{-b \pm \sqrt{\Delta}}{2a}$ are evaluated. A root is accepted if and only if the vertical coordinate $y(t) = o_y + t d_y$ satisfies $y_0 \leq y(t) \leq y_0 + h$. The surface normal is $\mathbf{N} = \text{normalize}([p_x - c_x, 0, p_z - c_z]^T)$.

3.7.4 Exact Code Citation

- **GPU Ray-Sphere Intersection:** `shaders/raytrace.frag`: lines 41--55
- **GPU Ray-Box Slab Intersection:** `shaders/raytrace.frag`: lines 58--84
- **GPU Ray-Cylinder Intersection:** `shaders/raytrace.frag`: lines 87--115
- **CPU Ray Tracer Equivalents:** `src/RayTracer.h`: lines 41--112

3.8 Parametric Curves and Sweep Mathematics

3.8.1 Catenary Sagging Formulation

The sacred *Shimenawa* straw ropes hanging from the Torii gate and the overhead street lantern spans exhibit natural gravitational catenary sag between two pole anchors at $x = -L$ and $x = +L$ with height y_{pole} and central dip s . The ideal hyperbolic catenary $y(x) = a \cosh(x/a) - a$ is accurately approximated using a computationally efficient second-degree parabola:

$$\tilde{x} = \frac{x}{L} \in [-1, 1], \quad y(x) = y_{\text{pole}} - s \cdot (1 - \tilde{x}^2) \quad (3.27)$$

The tangent vector along the curve used to orient sweeping frames is obtained by differentiation:

$$\frac{dy}{dx} = \frac{2s\tilde{x}}{L} \implies \mathbf{T}(x) = \text{normalize} \left(\begin{bmatrix} 1 \\ \frac{2s\tilde{x}}{L} \\ 0 \end{bmatrix} \right) \quad (3.28)$$

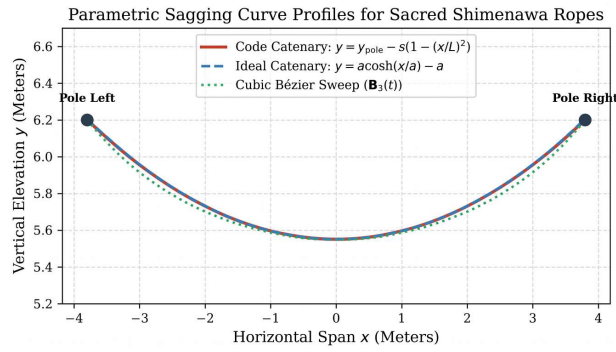


Figure 3.3: Comparison of the parabolic sagging curve against ideal hyperbolic catenary and cubic Bézier formulations.

3.8.2 Parametric Bézier Curves

Quadratic and Cubic Bézier curves are evaluated using Bernstein polynomials:

$$\mathbf{B}_2(t) = (1-t)^2 \mathbf{P}_0 + 2(1-t)t \mathbf{P}_1 + t^2 \mathbf{P}_2, \quad t \in [0, 1] \quad (3.29)$$

$$\mathbf{B}_3(t) = (1-t)^3 \mathbf{P}_0 + 3(1-t)^2 t \mathbf{P}_1 + 3(1-t)t^2 \mathbf{P}_2 + t^3 \mathbf{P}_3, \quad t \in [0, 1] \quad (3.30)$$

Their first derivatives yield the local curve tangent vector $\mathbf{T}(t) = \text{normalize}(\mathbf{B}'(t))$.

3.8.3 Bishop Frame Parallel Transport

Standard Frenet-Serret reference frames $(\mathbf{T}, \mathbf{N}, \mathbf{B})$ fail when a 3D spline curve undergoes inflection points ($\kappa = 0$), causing the normal vector to flip 180° and producing catastrophic pinching or twisting in swept tubes. To guarantee rotation-minimizing swept meshes, *Matsuri Nights* implements **Bishop frame parallel transport**:

Given an arbitrary initial normal $\mathbf{N}_0 \perp \mathbf{T}_0$, subsequent normals $\mathbf{N}_{i+1}$ along discretized curve points $\mathbf{P}_i$ are transported via the minimum-rotation axis $\mathbf{A} = \mathbf{T}_i \times \mathbf{T}_{i+1}$:

$$\theta_i = \arccos(\mathbf{T}_i \cdot \mathbf{T}_{i+1}), \quad \mathbf{N}_{i+1} = \text{Rotate}(\mathbf{N}_i, \theta_i, \text{normalize}(\mathbf{A})), \quad \mathbf{B}_{i+1} = \mathbf{T}_{i+1} \times \mathbf{N}_{i+1} \quad (3.31)$$

3.8.4 Parabolic Projectile Kinematics for Hopping Takoyaki Balls

Each of the 6 food balls on the Takoyaki stall griddle executes continuous jumping motion. The vertical hop elevation $\Delta y(t)$ is parameterized across period $T = 0.95$ s with apex height $h_{\text{max}} = 0.18$ m as:

$$\tau = \frac{t \pmod T}{T} \in [0, 1], \quad \Delta y(\tau) = 4h_{\text{max}} \cdot \tau \cdot (1 - \tau) \quad (3.32)$$

3.8.5 Exact Code Citation

- **Catenary Sagging Curve:** `src/Curves.h`: lines 613--625 in `Curves::createCatenaryRope()`
- **Bézier 2 & 3 Evaluation:** `src/Curves.h`: lines 37--81 in `Bezier2` and `Bezier3`
- **Bishop Frame Parallel Transport:** `src/Curves.h`: lines 162--215 in `Curves::computeBishopFrames()`
- **Takoyaki Projectile Motion:** `src/Objects.h`: lines 3125--3150 in `TakoyakiStall::update()`

4. Geometric Primitives and Swept Geometries

4.1 Geometric Primitive Factory Architecture

Rather than relying on static external OBJ or FBX model files, all geometry in *Matsuri Nights* is generated procedurally via mathematical functions defined in `src/Primitives.h` and `src/Curves.h`. This procedural architecture ensures that vertex normals, texture coordinates, and tangent frames are mathematically exact and eliminates external asset loading overhead.

4.2 Standard Geometric Primitives

Table 4.1 lists the fundamental primitives provided by `Primitives.h`, detailing their vertex layout, triangle counts, and procedural construction.

Primitive	Vertex Count	Index Count	Construction Methodology & Normal Layout
Cube (<code>createCube</code>)	24 vertices	36 indices (12 triangles)	6 independent quad faces (+Z, -Z, -X, +X, +Y, -Y). Each face possesses 4 unique vertices with duplicate corner positions to ensure sharp, unshared face normals ($\pm X, \pm Y, \pm Z$).
Cylinder (<code>createCylinder</code>)	102 vertices	288 indices (96 triangles)	Constructed with $S = 24$ radial segments. Comprises $(S + 1) \times 2 = 50$ side wall vertices with smooth radial normals $\mathbf{N} = [\cos \theta, 0, \sin \theta]^T$, 26 top disc cap vertices with $\mathbf{N} = [0, 1, 0]^T$, and 26 bottom cap vertices with $\mathbf{N} = [0, -1, 0]^T$.
Cone (<code>createCone</code>)	52 vertices	144 indices (48 triangles)	Side mantle evaluated over $S = 24$ segments with slanted outward normals $\mathbf{N} = \text{normalize}([r \cos \theta, r/h, r \sin \theta]^T)$, sharing apex vertex, plus a closed circular base cap disc.
Sphere (<code>createSphere</code>)	525 vertices	2880 indices (960 triangles)	UV sphere evaluated across $R = 20$ latitude rings and $S = 24$ longitude sectors. Radial outward normals $\mathbf{N} = \mathbf{P}/r$, with polar caps degenerating gracefully into triangle fans.
Plane (<code>createPlane</code>)	4 vertices	6 indices (2 triangles)	Horizontal quad spanning $[-w/2, +w/2] \times [-d/2, +d/2]$ with upward normal $\mathbf{N} = [0, 1, 0]^T$ and normalized UV mapping $[0, 1]^2$.

Table 4.1: Summary of Standard Geometric Primitives (`src/Primitives.h`).

4.3 Parametric Swept Tubes and Curved Architectural Beams

4.3.1 Generalized Swept Tubes via Bishop Frames

The engine provides a generalized swept surface generator, `Curves::createSweptTube`, which takes arbitrary 3D parametric trajectory functions $\mathbf{P}(t)$, tangent evaluators $\mathbf{T}(t)$, and variable radius functions $R(t)$. Using the Bishop frame parallel transport algorithm described in Section 3.31, a circle of radius $R(t_i)$ is swept across longitudinal slices:

$$\mathbf{V}_{i,j} = \mathbf{P}(t_i) + R(t_i) \cdot [\cos(\phi_j)\mathbf{N}_i + \sin(\phi_j)\mathbf{B}_i] \quad (4.1)$$

where $\phi_j = \frac{2\pi j}{S_{\text{rad}}}$ and surface normals correspond to $\mathbf{N}_{\text{surf}} = \cos(\phi_j)\mathbf{N}_i + \sin(\phi_j)\mathbf{B}_i$.

4.3.2 Curved Architectural Beams (*Kasagi* and *Shimaki*)

The prominent upward-flaring top beam (*Kasagi*) and sub-beam (*Shimaki*) of the Torii gate cannot be represented via straight cylinders or boxes. The engine implements `Curves::createCurvedBeam`, which sweeps an asymmetric trapezoidal cross-sectional profile along a cubic Bézier spine curve:

- **Cubic Bézier Spine:** Control points $\mathbf{P}_0 = [-3.8, 6.4, 0]$, $\mathbf{P}_1 = [-1.5, 6.15, 0]$, $\mathbf{P}_2 = [1.5, 6.15, 0]$, $\mathbf{P}_3 = [3.8, 6.4, 0]$, providing an authentic Japanese shrine curve with pronounced upward flare at the eaves (*Sori*).
- **Cross-Section:** 4 corner vertices defining a trapezoid wider at the top than the bottom, with chamfered rain gutters along the roof ridge.

4.3.3 Sacred Shimenawa Catenary Ropes

The sacred straw rope spanning the Torii gate pillars is generated via `Curves::createCatenaryRope` (444 vertices, 864 triangles). Using Equations 3.27 and 3.28, the rope maintains a natural gravitational sag with 36 longitudinal slices and 12 radial segments.

4.4 Organic Botanical and Human Anatomical Meshes

To avoid sterile blocky geometry, `src/Curves.h` implements specialized procedural synthesizers for botanical and human characters:

1. **Curved Cherry Blossom Petals** (`createCurvedPetalMesh`): Generates individual petals featuring gentle 3D concave cupping along the central midrib and a distinctive notched petal apex tip.
2. **Cherry Blossom Clusters** (`createSakuraBlossomLobe`): Assembles 5 radially arranged curved petals around a central pistil and stamen core, forming realistic blossom clusters.
3. **Pine Needle Fans** (`createPineNeedleClusterMesh`): Generates radial needle tufts for bonsai trees and planter boxes.
4. **Articulated Human Head** (`createHumanHeadMesh`): A contoured cranial mesh with tapered jawline, occipital curvature, and neck socket.
5. **Kimono Torso** (`createHumanTorsoMesh`): Models traditional Japanese festival attire with chest taper, overlapping kimono lapels (*Eri*), and a raised waist sash (*Obi*).
6. **Articulated Limbs** (`createArticulatedLimbMesh`): Tapered limb cylinders capped with rounded hemispheres representing shoulder, elbow, hip, and knee pivot sockets.
7. **Geta Sandal Foot** (`createGetaFootMesh`): Generates traditional elevated wooden clog sandals (*Geta*) featuring an elevated base plate, two vertical ground teeth (*Ha*), and an angled fabric toe strap (*Hanao*).

4.5 OpenGL Mesh Buffer Management

Each synthesized primitive is uploaded to GPU memory via the `Mesh` class (`src/Mesh.h`). Vertex attributes are interleaved into a single Vertex Buffer Object (VBO):

```
struct Vertex {
    glm::vec3 Position; // Offset 0, Size 12 bytes
    glm::vec3 Normal;   // Offset 12, Size 12 bytes
    glm::vec2 TexCoords; // Offset 24, Size 8 bytes
}; // Total stride: 32 bytes
```

The Vertex Array Object (VAO) binds the VBO and Element Buffer Object (EBO), configuring layout locations:

- **Location 0** (aPos): `glVertexAttribPointer(0, 3, GL_FLOAT, GL_FALSE, 32, (void*)0)`
- **Location 1** (aNormal): `glVertexAttribPointer(1, 3, GL_FLOAT, GL_FALSE, 32, (void*)12)`
- **Location 2** (aTexCoords): `glVertexAttribPointer(2, 2, GL_FLOAT, GL_FALSE, 32, (void*)24)`

Draw calls are dispatched via indexed rendering: `glDrawElements(GL_TRIANGLES, indexCount, GL_UNSIGNED_INT, 0)`.

5. Scene Graph Architecture and Hierarchy

5.1 Scene Graph Architectural Overview

A flat collection of isolated 3D meshes cannot model complex architectural compositions, moving limbs, or physical assemblies where components are anchored to moving parents. To solve this, *Matsuri Nights* implements an object-oriented hierarchical **Scene Graph** rooted at `SceneNode` (`src/SceneNode.h`).

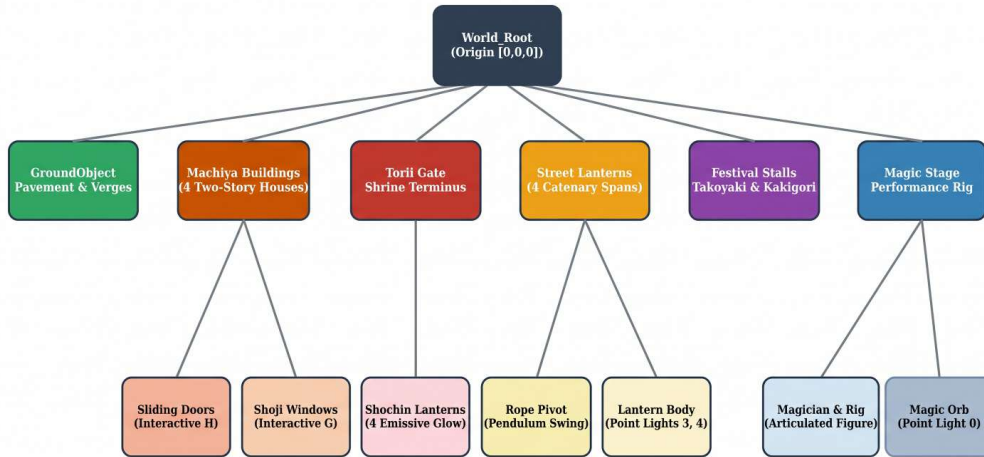


Figure 5.1: Hierarchical Scene Graph Tree illustrating the inheritance of spatial reference frames from `World.Root` to sub-assemblies.

5.2 The SceneNode Class Implementation

Each node in the scene graph encapsulates spatial, material, and topological state:

```
class SceneNode : public std::enable_shared_from_this<SceneNode> {
public:
    std::string name;
    Transform transform; // Local affine transform
    glm::mat4 worldMatrix{ 1.0f }; // Cascaded world transform
    glm::mat3 normalMatrix{ 1.0f }; // Inverse transpose for normals

    SceneNode* parent = nullptr; // Non-owning back-pointer to parent
    std::vector<std::shared_ptr<SceneNode>> children;

    const Mesh* mesh = nullptr; // Geometric primitive pointer
    glm::vec4 color{ 1.0f }; // Base diffuse tint
    bool isWindow = false; // Shoji paper transmission flag
    bool isSky = false; // Celestial background flag
    bool isEmissive = false; // Glowing lantern/firework flag
    glm::vec3 emissiveColor{ 0.0f };

    float shininess = 32.0f; // Blinn-Phong specular power
    float specularStrength = 0.5f;

    const Texture* texture = nullptr; // Diffuse map pointer
    float textureTiling = 1.0f;

    bool isStatic = false; // Caching optimization flag
    bool castShadow = true; // Shadow depth pass participation flag
};
```

5.2.1 Matrix Propagation and Dirty-Flag Optimization

Computing full matrix cascades for hundreds of nodes every frame incurs severe CPU cache overhead. `SceneNode` implements an aggressive dirty-flag propagation system (`updateWorldMatrix`):

1. **Local Transform Inspection:** `transform.checkDirty()` returns true if position, rotation, or scale changed since the previous evaluation.
2. **Parent Change Detection:** The node verifies whether the incoming `parentMatrix` differs from `lastParentMatrix`.
3. **Early Exit for Static Subtrees:** When `isStatic` is set to true and neither the node nor its ancestors mutated, matrix re-evaluation and child traversal are entirely skipped.

5.2.2 RenderContext and State Cache Minimization

Calling `glGetUniformLocation` during the render loop incurs expensive OpenGL driver hash table lookups. To prevent this, `SceneNode.h` introduces `RenderContext`:

- Uniform locations are queried and stored once during initialization (`ctx.init(shader)`).
- Uniform values (`lastShininess`, `lastSpecularStrength`, `lastTexture`) are cached within the context. Redundant uniform setter calls are bypassed unless the material properties actually change between nodes.

5.3 Catalog of the 17 Scene Entities

The complete virtual festival environment is constructed from 17 architectural and animated entities initialized in `Scene::buildScene()` (`src/Scene.h`):

1. **Ground** (`GroundObject`): A 9m wide, 80m long stone cobblestone street flanked by grass verges and granite curbstones.
2. **Machiya L1 & L2** (`MachiyaBuilding`): Two traditional 2-story townhouses on the left side of the street featuring sliding Shoji entrance doors, sliding windows, interior Tatami rooms, and curved tile eaves.
3. **Machiya R1 & R2** (`MachiyaBuilding`): Symmetrical 2-story townhouses lining the right side of the street.
4. **Torii Gate** (`ToriiGate`): Grand vermilion shrine gate at street terminus ($z = -32.0\text{m}$) with curved Kasagi beam, sacred Shimenawa catenary rope, Shide paper streamers, hanging Chochin lanterns, bracket lanterns, and stone Toro lanterns.
5. **Sakura Trees** (`SakuraTree`): Weeping cherry blossom trees with cubic Bézier trunks, 5 main boughs, 15 secondary branches, clustered blossom lobes, and 12 animated falling petals.
6. **Street Lantern Spans** (`StreetLanternSpan`): 4 overhead catenary spans crossing the street ($z = 11.0, 1.0, -9.0, -19.0\text{m}$), each carrying hanging paper lanterns with swinging physics.
7. **Takoyaki Stall** (`TakoyakiStall`): Street food stall featuring timber counters, Noren fabric banners, hot cast-iron griddle plate, and 6 animated hopping takoyaki balls.
8. **Kakigori Stall** (`KakigoriStall`): Shaved ice dessert stall with vintage hand-cranked ice shaving machine, rotating flywheel, translucent ice block, and colorful dessert bowls.
9. **Vendor Figures** (`VendorFigure`): Articulated stall chefs wearing festival kimonos, aprons, and headbands (*Hachimaki*).
10. **Magic Stage** (`MagicStage`): Elevated performance stage ($z = -20.0\text{m}$) with dark timber planks and rear decorative curtain frame.
11. **Magician** (`Magician`): Articulated human magician in black Haori jacket with gesturing arms and wand.
12. **Vanishing Box** (`VanishingBoxTrick`): Golden illusion chest executing a 5-phase vanishing and scale-to-zero trick.
13. **Spotlight Rig** (`SpotlightRig`): Overhead theatrical truss ($y = 7.5\text{m}$) carrying a rotating spotlight housing that tracks the stage.
14. **Audience Group** (`AudienceGroup`): 6 seated spectators wearing varied kimonos, head turns, and cheering animations.
15. **Crowd Group** (`CrowdGroup`): 4 walking pedestrians with swinging limbs, geta sandals, and head bobbing.
16. **Fireworks System** (`FireworkSystem`): Multi-rocket particle launcher with launch trails, apex bursts, and expanding sparks.
17. **Sky Dome** (`SkyDome`): Infinite celestial sphere ($R = 250\text{m}$) tracking camera position, rendering the dynamic sun, moon, and procedural stars.

6. Illumination, Materials, and Shading Pipeline

6.1 The 16-Light Dynamic Optical Architecture

The illumination pipeline in *Matsuri Nights* models the complex nocturnal atmosphere of an illuminated festival street. Shaders operate in forward rendering with **16 active dynamic light sources**: 1 Directional Light (Sun/Moon), 14 Point Lights, and 1 Theatrical Spotlight. Table 6.1 provides the complete technical specification of every light source.

Light ID	Type	Physical Entity	Diffuse Color	Specular Color	Attenuation / Coupling
DirLight	Directional	Celestial Sun / Moon	Day: (0.85, 0.82, 0.76) Night: (0.20, 0.25, 0.38)	Day: (0.60, 0.60, 0.55) Night: (0.30, 0.35, 0.45)	Vector (0.4, -0.85, -0.5) $\rightarrow$ (-0.35, -0.75, 0.4). lerps $\rightarrow$ Casts PCF shadows.
Point 0	Point	Magic Orb	(0.35, 0.85, 1.00)	(0.50, 0.90, 1.00)	$k_l = 0.14, k_q = 0.07$. Tracks Magician hand bone.
Point 1	Point	Takoyaki Stall Eaves	(1.00, 0.60, 0.22)	(1.00, 0.70, 0.30)	$k_l = 0.10, k_q = 0.045$. Pos: (-5.2, 2.85, 6.0).
Point 2	Point	Kakigori Stall Eaves	(0.35, 0.90, 1.00)	(0.50, 0.95, 1.00)	$k_l = 0.10, k_q = 0.045$. Pos: (5.2, 2.85, 6.0).
Point 3	Point	Street Lantern Span 1	(1.00, 0.55, 0.20)	(1.00, 0.65, 0.25)	$k_l = 0.09, k_q = 0.032$. Tracks swinging lantern body.
Point 4	Point	Street Lantern Span 3	(1.00, 0.55, 0.20)	(1.00, 0.65, 0.25)	$k_l = 0.09, k_q = 0.032$. Tracks swinging lantern body.
Point 5	Point	Fireworks Sky Burst	Burst RGB $\times 1.8$	Burst RGB $\times 1.5$	$k_l = 0.04, k_q = 0.009$. Active during rocket apex burst.
Point 6	Point	Machiya L1 Floor 1 Lamp	(1.45, 1.20, 0.75)	(0.90, 0.80, 0.50)	$k_l = 0.08, k_q = 0.022$. Pos: (-10.8, 2.2, 17.1).
Point 7	Point	Machiya L1 Floor 2 Lamp	(1.35, 1.10, 0.70)	(0.80, 0.70, 0.45)	$k_l = 0.08, k_q = 0.022$. Pos: (-11.0, 6.0, 17.0).
Point 8	Point	Machiya R1 Floor 1 Lamp	(1.45, 1.20, 0.75)	(0.90, 0.80, 0.50)	$k_l = 0.08, k_q = 0.022$. Pos: (10.8, 2.2, 14.9).
Point 9	Point	Machiya R1 Floor 2 Lamp	(1.35, 1.10, 0.70)	(0.80, 0.70, 0.45)	$k_l = 0.08, k_q = 0.022$. Pos: (11.0, 6.0, 15.0).
Point 10	Point	Machiya L2 Floor 1 Lamp	(1.35, 1.10, 0.70)	(0.80, 0.70, 0.45)	$k_l = 0.08, k_q = 0.022$. Pos: (-10.8, 2.2, -4.9).
Point 11	Point	Machiya R2 Floor 1 Lamp	(1.35, 1.10, 0.70)	(0.80, 0.70, 0.45)	$k_l = 0.08, k_q = 0.022$. Pos: (10.8, 2.2, -7.1).
Point 12	Point	Torii Gate Left Lantern	(1.50, 1.05, 0.50)	(1.30, 1.00, 0.55)	$k_l = 0.07, k_q = 0.018$. Pos: (-2.8, 6.8, -31.5).
Point 13	Point	Torii Gate Right Lantern	(1.50, 1.05, 0.50)	(1.30, 1.00, 0.55)	$k_l = 0.07, k_q = 0.018$. Pos: (2.8, 6.8, -31.5).
SpotLight	Spotlight	Stage Theatrical Truss	(1.50, 1.35, 1.10)	(1.20, 1.10, 0.90)	$\theta_{in} = 15^\circ, \theta_{out} = 23^\circ$. Tracks Magician.

Table 6.1: Comprehensive Catalog of All 16 Active Dynamic Light Sources (`shaders/basic.frag`).

6.2 Shading Modes Comparison

To demonstrate how individual optical components contribute to the final image, pressing the **P** key cycles between three shading modes:

- Mode 0: Blinn-Phong Illumination** ($I = I_a + I_d + I_s$): Full physical approximation combining ambient floor bounce, Lambertian diffuse scattering, and specular highlights calculated via halfway vector **H**.
- Mode 1: Diffuse-Only** ($I = I_a + I_d$): Specular highlights are zeroed out, simulating purely matte, rough Lambertian surfaces.
- Mode 2: Ambient-Only** ($I = I_a$): Both diffuse scattering and specular reflection are removed, illustrating the unoccluded baseline indirect illumination.

6.3 Day / Night Sky Dome and Atmospheric Transition

Pressing the **N** key smoothly drives the state variable `dayNightFactor` between 0.0 (Daylight) and 1.0 (Festival Night). Figure 6.2 plots the modulation of celestial and artificial light intensities across this transition.



Figure 6.1: Visual comparison of the three runtime shading modes accessible via Key P.

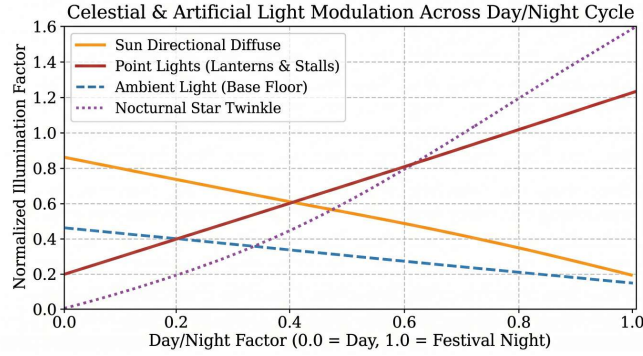


Figure 6.2: Continuous light modulation curves across the Day/Night state transition.

6.3.1 Procedural Celestial Bodies

The celestial sphere (`isSky == true`) evaluates view direction $\mathbf{V} = \text{normalize}(\mathbf{p}_{\text{frag}} - \mathbf{p}_{\text{cam}})$:

- **Celestial Sun (Daylight):** Emits a sharp core disc when $\mathbf{V} \cdot \mathbf{D}_{\text{sun}} > 0.9972$ with a wide exponential atmospheric corona bloom $(\mathbf{V} \cdot \mathbf{D}_{\text{sun}})^{64} \times 1.10$.
- **Celestial Moon (Nighttime):** Silvery-white disc with a crater modulation function $\sin(16u) \cos(18v) \times 0.12$ and an ethereal bluish outer halo.
- **Twinkling Starfield:** Evaluated across two spatial grid frequencies ($160\times$ and $320\times$). A pseudo-random hash generator extracts star seeds, animating individual brightnesses with sinusoidal twinkle frequencies:

$$I_{\text{star}} = \text{smoothstep}(r_{\text{star}}, 0, d) \cdot [0.65 + 0.35 \sin(\omega t + \phi)] \cdot \text{pow}(\text{dayNightFactor}, 1.25) \quad (6.1)$$

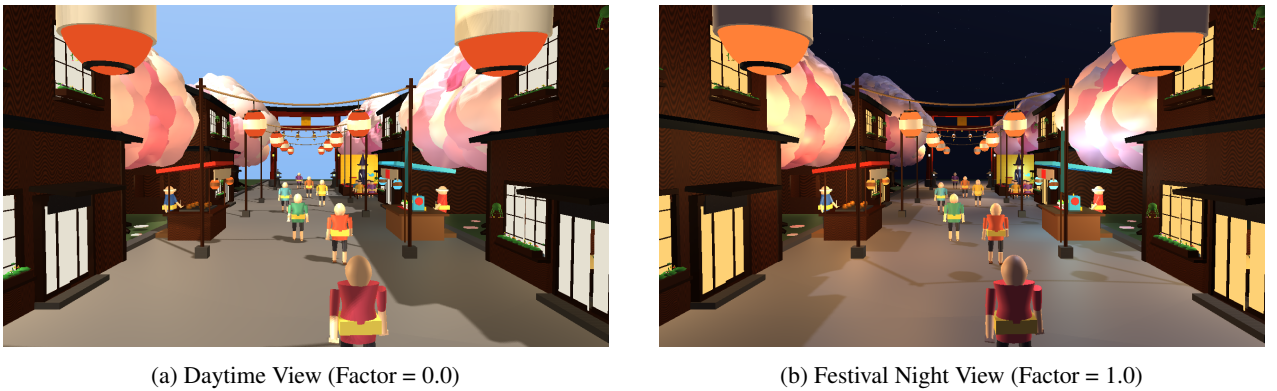


Figure 6.3: Comparison of the festival street under bright daytime and nocturnal illumination.

6.4 Specialized Material Shading Models

1. **Shoji Paper Translucency (`isWindow`):** Traditional sliding paper screens transmit interior and exterior light. In daytime, screens appear as crisp neutral parchment; at night, indoor lamps (Point Lights 6–11) back-illuminate screens into glowing warm amber radiators (1.0, 0.82, 0.45).
2. **Emissive Materials (`isEmissive`):** Chochin paper lanterns and firework sparks emit radiant self-illumination independent of directional shadow occlusion.

3. **Indoor Ambient Bounce:** To prevent interiors from becoming pitch-black cavities when doors or windows are open, the shader blends a warm ambient bounce color:

$$\mathbf{I}_{\text{indoorBounce}} = \text{mix} \left([0.48, 0.45, 0.40]^T, [0.18, 0.16, 0.14]^T, \text{dayNightFactor} \right) \cdot \mathbf{k}_d \quad (6.2)$$

7. Realistic Soft Shadow Mapping

7.1 Two-Pass Shadow Mapping Architecture

Hard shadows produce unrealistic, razor-sharp silhouettes and severe aliasing artifacts. To achieve visual fidelity, *Matsuri Nights* incorporates an optimized two-pass **Percentage-Closer Filtering (PCF) Soft Shadow Mapping** system.

7.1.1 Pass 1: Directional Light Depth Generation

In the first render pass (`Scene::renderShadowDepth`):

1. An offscreen Framebuffer Object (FBO), `depthMapFBO`, is bound with dimensions 2048×2048 .
2. A single 2D depth texture, `depthMap`, formatted as `GL_DEPTH_COMPONENT`, is attached. The texture wrap modes are configured to `GL_CLAMP_TO_BORDER` with a border color of $[1.0, 1.0, 1.0, 1.0]^T$, ensuring fragments outside the light frustum are never falsely shadowed.
3. **Peter-Panning Mitigation:** Hardware face culling is switched to front faces (`glCullFace(GL_FRONT)`). Consequently, the depth buffer captures back faces, preventing self-shadowing light leakage between touching geometry.
4. The directional light's orthographic projection matrix and view matrix are computed:

$$\mathbf{P}_{\text{ortho}} = \text{ortho}(-35.0, 35.0, -25.0, 25.0, 1.0, 90.0) \quad (7.1)$$

$$\mathbf{V}_{\text{light}} = \text{lookAt}\left(-\mathbf{D}_{\text{light}} \times 40.0\text{m}, \begin{bmatrix} 0 \\ 0 \\ -5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) \quad (7.2)$$

$$\mathbf{M}_{\text{lightSpace}} = \mathbf{P}_{\text{ortho}} \cdot \mathbf{V}_{\text{light}} \quad (7.3)$$

5. The minimal depth shader (`shaders/shadow_depth.vert`) transforms all non-emissive meshes into light clip space.

7.1.2 Pass 2: Shading Pass with PCF Filtering

During the primary forward rendering pass, the generated depth map is bound to `GL_TEXTURE1`. The fragment shader (`shaders/basic.frag: calculateShadow`) computes projected light-space coordinates:

$$\mathbf{p}_{\text{proj}} = \left(\frac{\mathbf{p}_{\text{lightSpace}} \cdot \mathbf{XYZ}}{\mathbf{p}_{\text{lightSpace}} \cdot \mathbf{W}} \right) \cdot 0.5 + 0.5 \quad (7.4)$$

If coordinates fall outside $[0, 1]^3$, shadow occlusion is set to 0.0.

7.2 Artifact Mitigation Strategies

7.2.1 Adaptive Slope-Scale Depth Bias

Flat bias constants fail on steep surfaces, generating either shadow acne (bias too small) or light detachment/Peter-Panning (bias too large). Equation 3.19 dynamically scales bias based on the angle between the surface normal $\mathbf{N}$ and incident light direction $\mathbf{L}$:

```
float bias = max(0.0035 * (1.0 - dot(normal, lightDir)), 0.0006);
```

7.2.2 16-Sample PCF Convolution

Rather than performing a binary depth comparison, the shader samples a 4×4 disc kernel centered around the projected texel with texel spacing $\Delta = \frac{1.2}{2048.0}$:

```
float shadow = 0.0;
const vec2 texelSize = vec2(1.0 / 2048.0);
for(int x = -1; x <= 2; ++x) {
    for(int y = -1; y <= 2; ++y) {
        float pcfDepth = texture(shadowMap, projCoords.xy + vec2(x, y) * texelSize * 1.2).r;
        shadow += (projCoords.z - bias > pcfDepth) ? 1.0 : 0.0;
    }
}
shadow /= 16.0;
```

This filtering produces soft contact penumbrae under Machiya roof eaves, Torii crossbeams, and street stalls.

7.3 Visual Verification

Pressing the **V** key toggles shadow mapping. Figure 7.1 illustrates the profound visual improvement achieved by 16-sample PCF soft shadows over unshadowed rendering.



(a) Shadows Disabled (Flat unshadowed lighting)



(b) 16-Sample PCF Soft Shadows Enabled

Figure 7.1: Visual impact of realistic 16-sample PCF soft shadow mapping accessible via Key V.

8. Texturing System and Procedural Synthesis

8.1 Procedural Texture Generation Architecture

To achieve complete self-containment without dependency on external image downloads, *Matsuri Nights* implements an autonomous procedural texture synthesizer within `src/TextureGenerator.h`. At engine startup, if static BMP files are missing from `assets/textures/`, the generator programmatically synthesizes eight 24-bit uncompressed BMP textures (256×256 pixels, exactly 196,662 bytes each).

8.1.1 Raw 24-Bit BMP Image Exporter

The function `TextureGenerator::writeBMP24` constructs standard Windows BMP files by assembling the 14-byte `BITMAPFILEHEADER` and 40-byte `BITMAPINFOHEADER`:

$$\text{RowBytes} = \text{ceil}\left(\frac{\text{width} \times 3}{4}\right) \times 4 = (256 \times 3 + 3) \& (\sim 3) = 768 \text{ bytes} \quad (8.1)$$

Pixels are transformed from standard RGB into the BGR byte layout expected by the BMP format, handling 4-byte row alignment padding.

8.2 The Eight Procedural Texture Synthesizers

Table 8.1 summarizes the procedural generation algorithms for the eight scene materials.

Texture Name	Mapped Geometry	Mathematical Synthesis Algorithm
Wood Timber (<code>wood_timber.bmp</code>)	Machiya timber framing, Torii crossbeams, stall counters	Combines longitudinal high-frequency sine waves with multi-octave pseudo-noise to model authentic wood grain fibers and ring density variations.
Roof Tiles (<code>roof_tiles.bmp</code>)	Machiya primary & eave roofs, stall canopies	Models traditional curved Japanese ceramic tiles (<i>Kawara</i>) using periodic vertical cylindrical profile shading overlaid with horizontal tile overlapping lips.
Stone Pavement (<code>stone_pavement.bmp</code>)	Ground street pavement, building curbstones	Generates staggered rectangular granite cobblestones with deep mortar grooves, perturbed with surface roughness noise.
Lantern Paper (<code>lantern_paper.bmp</code>)	Chochin paper lanterns, Andon floor lamps	Simulates translucent handmade mulberry paper (<i>Washi</i>) with horizontal bamboo rib line banding and subtle fiber density variations.
Tatami Cloth (<code>tatami_cloth.bmp</code>)	Machiya interior ground and upper floor decks	Models woven rush grass Tatami mat weave patterns with dark fabric border binding bands along mat edges.
Gold Leaf (<code>gold_leaf.bmp</code>)	Vanishing illusion box, Torii plaque inscription	High-frequency metallic noise producing glittering specular micro-facets of hammered decorative gold leaf.
Sakura Bark (<code>sakura_bark.bmp</code>)	Cherry blossom tree trunk and main boughs	Deep charcoal-brown base color perturbed by characteristic horizontal lenticel breathing pores and furrowed bark cracks.
Takoyaki Batter (<code>takoyaki_food.bmp</code>)	Hopping takoyaki octopus food balls	Golden-brown fried batter gradient decorated with green seaweed flakes (<i>Aonori</i>) and dark brown Worcestershire sauce drizzle stripes.

Table 8.1: Catalog of Procedural 24-bit BMP Textures (`src/TextureGenerator.h`).

8.3 Texture Mapping and GPU Sampling Pipeline

Textures are managed via the `Texture` class (`src/Texture.h`):

- **Filtering Modes:** Minification uses `GL_LINEAR_MIPMAP_LINEAR` (trilinear filtering with mipmaps); magnification uses `GL_LINEAR`.
- **Wrap Modes:** S and T coordinates are configured to `GL_REPEAT`, allowing arbitrary texture repetition across large surface areas (such as the 80m stone street).
- **Texture Tiling:** Each `SceneNode` maintains a `textureTiling` uniform scalar, scaling coordinates ($u' = u \cdot \text{tiling}$, $v' = v \cdot \text{tiling}$) in the fragment shader.

8.4 Visual Comparison: Textures Enabled vs. Disabled

Pressing the **X** key toggles diffuse texture maps globally, revealing the raw underlying material colors and geometric meshes.



(a) Textures Disabled (Raw geometric material colors)



(b) Textures Enabled (Diffuse maps active)

Figure 8.1: Visual impact of diffuse texture mapping accessible via Key X.

9. Physics, Animations, and Relative Motion

9.1 Overview of Kinematic Systems

A static 3D world fails to convey the vitality of a Japanese festival. *Matsuri Nights* incorporates six interconnected kinematic and physics animation systems. These systems specifically demonstrate **motion relative to another object's reference frame**, hierarchical articulation, and continuous collision detection.

9.2 System 1: Street Lantern Harmonic Pendulum Swing

9.2.1 Hierarchical Formulation

Overhead street lanterns are assembled from two hierarchical nodes:

1. **Parent Anchor Node** (`ropePivot`): Attached to the catenary suspension cable at $(x_{\text{span}}, y_{\text{cable}}, z_{\text{span}})$. Its local transformation contains zero translation but an oscillating Z-axis Euler rotation:

$$\theta_{\text{pivot}}(t) = \theta_0 \cdot \sin(\omega_{\text{swing}}t + \phi_k) \quad (9.1)$$

where amplitude $\theta_0 = 12.0^\circ$, frequency $\omega_{\text{swing}} = 1.85 \text{ rad/s}$, and ϕ_k is a staggered phase offset.

2. **Child Body Node** (`lanternBody`): Positioned at a negative vertical offset along local Y ($y_{\text{local}} = -1.2 \text{ m}$). As the parent rotates, the child sweeps through a circular pendulum arc in world space:

$$\mathbf{p}_{\text{lantern}}(t) = \mathbf{M}_{\text{pivot}}(t) \cdot \begin{bmatrix} 0 \\ -1.2 \\ 0 \\ 1 \end{bmatrix} \quad (9.2)$$

9.2.2 Coupling to Dynamic Moving Lights

Point Lights 3 and 4 query `lanternBody->getWorldPosition()` every frame (`src/Scene.h: lines 813--824`). Consequently, the illumination cast upon the cobblestone street and house facades swings in real time with the lanterns.

9.3 System 2: Magic Orb Orbital Hierarchy

9.3.1 Nested Reference Frame Hierarchy

The theatrical stage features an articulated magician figure performing with a glowing mystical orb. The orb is not animated in world coordinates; rather, it is anchored six levels deep within the magician's skeletal hierarchy:

$$\text{World} \rightarrow \text{Stage} \rightarrow \text{Magician Root} \rightarrow \text{Torso} \rightarrow \text{Right Shoulder} \rightarrow \text{Hand Bone} \rightarrow \text{Magic Orb} \quad (9.3)$$

Relative to the moving hand bone coordinate frame, the orb executes harmonic elliptical orbital motion:

$$\begin{aligned} x_{\text{local}}(t) &= R_{\text{orbit}} \cdot \cos(\omega_{\text{orb}}t) \\ z_{\text{local}}(t) &= R_{\text{orbit}} \cdot \sin(\omega_{\text{orb}}t) \\ y_{\text{local}}(t) &= y_0 + A_{\text{levitate}} \cdot \sin(2\omega_{\text{orb}}t) \end{aligned} \quad (9.4)$$

where $R_{\text{orbit}} = 0.45 \text{ m}$, $\omega_{\text{orb}} = 2.4 \text{ rad/s}$, and $A_{\text{levitate}} = 0.12 \text{ m}$. Point Light 0 tracks the orb's world position, illuminating the magician's face, stage backdrop, and audience spectators with pulsing cyan radiance.

9.4 System 3: Takoyaki Projectile Hops and Kakigori Shaver Wheel

9.4.1 Takoyaki Food Stall Kinematics

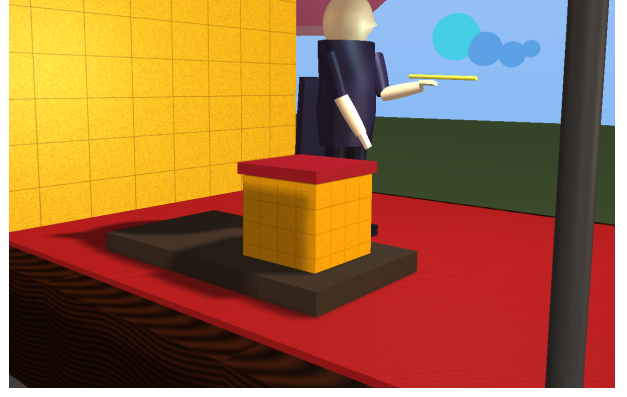
Each of the six individual octopus balls on the hot cast-iron griddle executes continuous jumping motion. Using Equation 3.32, vertical displacement follows a parabolic trajectory ($h_{\text{max}} = 0.18 \text{ m}$, $T = 0.95 \text{ s}$) while rotating continuously about the Y-axis ($360^\circ/\text{s}$).

9.4.2 Kakigori Shaved Ice Machine Kinematics

The vintage shaved ice machine features an iron flywheel crank rotating at $\omega_{\text{flywheel}} = 3.6 \text{ rad/s}$, geared to an internal ice block blade. On the counter, six colorful shaved ice bowls rotate synchronously.



(a) Magician Figure with Orbiting Cyan Orb



(b) Vanishing Box Illusion Trick



(c) Takoyaki Food Stall with Hopping Balls



(d) Kakigori Shaved Ice Dessert Stall

Figure 9.1: Theatrical magic show and culinary food stall animated assemblies.

9.5 System 4: Vanishing Box 5-Phase Illusion Sequence

The vanishing box trick (`src/Objects.h: lines 4246--4355`) operates via a state machine tracked by `stateTimer`:

- **Phase 1 (0.0s – 2.0s) — Levitation:** Golden box hovers upward from the table ($\Delta y = 0.35\text{ m}$) using a smooth cubic ease-in-out curve.
- **Phase 2 (2.0s – 3.2s) — Rapid Spin:** Box accelerates spin about the vertical axis ($\omega_y = 720^\circ/\text{s}$).
- **Phase 3 (3.2s – 4.2s) — Vanishing Scale:** Non-uniform scale collapses to zero:

$$S(t) = S_0 \cdot (1.0 - \tau^3), \quad \tau = \frac{t - 3.2}{1.0} \quad (9.5)$$

- **Phase 4 (4.2s – 6.0s) — Absence:** Box is completely invisible (`visible = false`); the magician gestures to the empty table.
- **Phase 5 ($> 6.0\text{s}$) — Replay:** Cycle resets or can be triggered at any time via Key M.

9.6 System 5: Hierarchical Pedestrian Crowd Walking

The pedestrian crowd figures walking down the street possess a 10-bone hierarchical character rig:

- **Linear Translation:** The root node advances down the street at $v_{\text{walk}} = 1.15\text{ m/s}$. When $z < -30.0\text{ m}$, the walker wraps around to $+28.0\text{ m}$.
- **Counter-Phase Hip Swings:** Left and right hip joints swing out of phase: $\theta_{\text{hip, L}} = \theta_{\text{leg}} \sin(\omega t)$, $\theta_{\text{hip, R}} = -\theta_{\text{leg}} \sin(\omega t)$ ($\theta_{\text{leg}} = 26.0^\circ$).
- **Realistic Knee Flexion:** Knees only bend backwards during the swing phase:

$$\theta_{\text{knee}} = \max(0.0, -\theta_{\text{hip}} \cdot 0.85) \quad (9.6)$$

- **Counter-Phase Arm Swings:** Left arm swings opposite to left leg to balance angular momentum ($\theta_{\text{arm}} = 18.0^\circ$).
- **Torso Bobbing:** Vertical bobbing simulates the gait cycle: $y = y_0 + 0.04 \cdot |\sin(\omega t)|$.

9.7 System 6: Interactive Sliding Shoji Doors and Windows

9.7.1 Proximity Detection and Smoothstep Sliding

When the camera approaches within 5.0 meters of any Machiya building entrance, pressing **H** toggles the front Shoji entrance door, while pressing **G** toggles the sliding windows. The sliding progress $\tau \in [0, 1]$ transitions smoothly across $T = 0.65$ s using cubic Hermite interpolation:

$$\text{smoothstep}(\tau) = 3\tau^2 - 2\tau^3, \quad x(t) = x_{\text{closed}} + \Delta x_{\text{max}} \cdot \text{smoothstep}(\tau) \quad (9.7)$$

where max doorway travel $\Delta x_{\text{max}} = 1.45$ m and window travel is 0.90 m.



(a) Pedestrian Crowd Walking Limb Cycles



(b) Apex Fireworks Sky Burst Particle Flash



(c) Exterior View of Townhouses L1 & L2



(d) Interior View through Open Shoji Door

Figure 9.2: Dynamic crowd pedestrians, sky fireworks, and interactive Machiya townhouses.

9.7.2 Continuous Collision Detection and Doorway Portals

Pressing the **B** key toggles between **Walk Mode** (solid walls and collision active) and **Noclip Mode** (free flying). In Walk Mode:

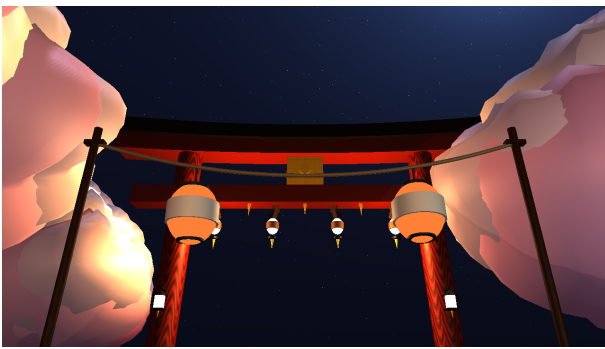
1. Bounding boxes encase Machiya foundation walls ($X \in [-14.5, -6.5]$, $Z \in [11.5, 20.5]$).
2. **Dynamic Doorway Portal:** A passage window is defined at $Z \in [-2.05, +0.25]$ m and $Y \in [0.0, 2.55]$ m. If `isDoorOpen == false`, the door acts as an impenetrable barrier, clamping camera position. When `isDoorOpen == true`, the portal opens, allowing the player to freely walk from the outdoor street into the interior living room.
3. **14-Step Staircase Elevation:** Inside the house, walking onto the staircase footprint smoothly interpolates player elevation Y from ground floor level (0.2 m) to the second floor deck (4.3 m).

9.8 Falling Sakura Petals and Fireworks Particle Physics

- **Cherry Blossom Petal Drift** (`src/Objects.h`: lines 2720--2770): Twelve animated petal nodes execute downward gravitational descent coupled with lateral sinusoidal wind swaying:

$$x(t) = x_0 + A_{\text{wind}} \sin(\omega_w t + \phi), \quad y(t) = y_0 - v_{\text{fall}} t \pmod{H_{\text{canopy}}} \quad (9.8)$$

- **Fireworks Particle Rockets** (`src/Objects.h`: lines 4917--5110): Pressing **F** launches an upward rocket with velocity $v_y = 28$ m/s. Upon reaching apex altitude ($y \approx 45$ m), the rocket explodes into a spherical cloud of expanding colored spark particles affected by air drag and gravity. Point Light 5 flashes at the burst center, casting dramatic transient illumination across the town.



(a) Grand Torii Shrine Gate with Lanterns



(b) Sakura Blossom Tree with Falling Petals

Figure 9.3: Cultural landmarks: Grand Torii shrine gate and cherry blossom tree.

10. Advanced Extensions: Dual Ray Tracing Engines

10.1 Motivation for Ray Tracing in OpenGL

Standard rasterization projects 3D polygons onto a 2D screen independently, calculating illumination on a strictly local basis. While shadow mapping enables approximate occlusion, rasterization inherently struggles with **recursive specular reflections** (mirror surfaces), true physical refractions, and analytic curve silhouettes. To explore both ends of the rendering spectrum, *Matsuri Nights* incorporates **two independent Whitted ray-tracing engines**:

1. A **Real-Time GPU Whitted Ray Tracer** running inside a full-screen fragment shader (`shaders/raytrace.frag`) toggled interactively via Key **Z**.
2. A **Multi-Threaded CPU Snapshot Ray Tracer** implemented in `src/RayTracer.h` running across all available CPU cores, triggered via Key **F9**.

10.2 Real-Time GPU Ray Tracer Architecture

10.2.1 Full-Screen Quad Setup

When ray-tracing mode is toggled via Key **Z**, the rasterization pipeline bypasses traditional geometry draw calls. A full-screen quad covering Normalized Device Coordinates (NDC) $[-1, 1]^2$ is rasterized:

```
// shaders/raytrace.vert
#version 330 core
layout (location = 0) in vec2 aPos;
out vec2 TexCoords;
void main() {
    TexCoords = (aPos + 1.0) * 0.5;
    gl_Position = vec4(aPos, 0.0, 1.0);
}
```

10.2.2 Per-Pixel Primary Camera Ray Generation

In the fragment shader (`shaders/raytrace.frag`), each pixel reconstructs its normalized screen space coordinates $(u_{\text{ndc}}, v_{\text{ndc}}) \in [-1, 1]^2$. Given camera orthonormal basis $\{\mathbf{F}, \mathbf{R}, \mathbf{U}\}$, field-of-view θ_{fov} , and aspect ratio a :

$$\mathbf{r}_{\text{dir}} = \text{normalize} \left(\mathbf{F} + \left(u_{\text{ndc}} \cdot a \cdot \tan \frac{\theta_{\text{fov}}}{2} \right) \mathbf{R} + \left(v_{\text{ndc}} \cdot \tan \frac{\theta_{\text{fov}}}{2} \right) \mathbf{U} \right) \quad (10.1)$$

10.2.3 Analytical Scene Representation

The festival street geometry is mapped analytically into geometric primitives:

- **Spheres:** Street lanterns, floating magic orb, firework particles, and takoyaki food balls evaluated via Equation 3.23.
- **AABB Boxes:** Townhouse foundations, walls, timber posts, stage platform, and vendor counters evaluated via the slab method (Equation 3.25).
- **Vertical Cylinders:** Torii gate pillars and street poles evaluated with height clamping.
- **Ground Plane:** Infinite street plane at $y = 0$.

10.2.4 Recursive Mirror Reflections and Hard Shadow Rays

For each hit point $\mathbf{p}_{\text{hit}}$:

1. **Shadow Rays:** A secondary ray is cast toward each active light source $\mathbf{r}_{\text{shadow}} = \mathbf{p}_{\text{hit}} + \epsilon \mathbf{N} + t \mathbf{L}$. If any occluder is struck between $\mathbf{p}_{\text{hit}}$ and the light source, direct diffuse and specular components are zeroed out.
2. **Recursive Reflection Rays:** If the hit material possesses non-zero reflectivity ($R > 0.02$) and the bounce count is below `uMaxBounces = 3`, a reflection ray is generated:

$$\mathbf{r}_{\text{refl}} = \text{reflect}(\mathbf{r}_{\text{dir}}, \mathbf{N}) = \mathbf{r}_{\text{dir}} - 2(\mathbf{r}_{\text{dir}} \cdot \mathbf{N})\mathbf{N} \quad (10.2)$$

The ray origin is offset by $+0.003\mathbf{N}$ to prevent self-intersection. Specular mirror reflections are vividly visible on the metallic gold vanishing box ($R = 0.85$), the magic orb ($R = 0.90$), and the polished stage floor ($R = 0.40$).

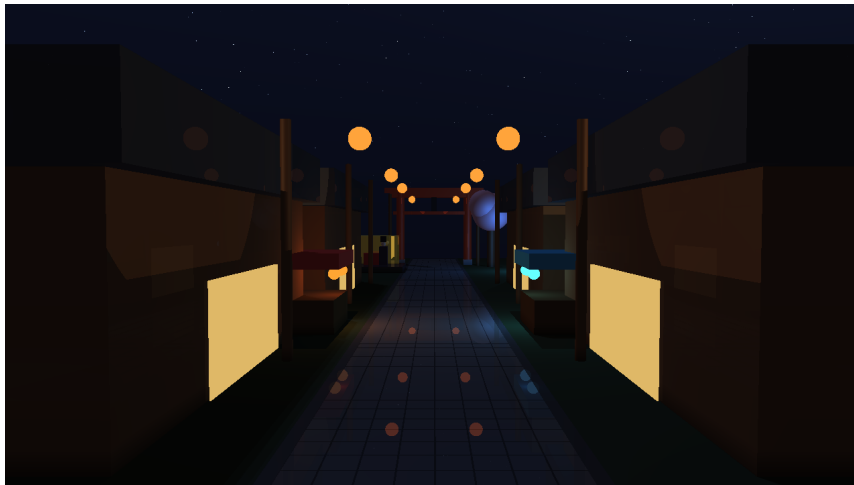


Figure 10.1: Real-time GPU Whitted ray tracer active in viewport, showing per-pixel analytical primary rays, hard shadows, and recursive mirror reflections.

10.3 Multi-Threaded CPU Snapshot Ray Tracer

10.3.1 Thread Pool Parallelization

Pressing Key **F9** activates the CPU ray tracer (`src/RayTracer.h: renderSnapshot()`):

- The system queries available hardware concurrency via `std::thread::hardware_concurrency()` (typically 8 to 16 threads).
- The image buffer (1280×720) is partitioned into equal horizontal scanline stripes (Y slabs).
- Each thread renders its assigned rows into a shared contiguous 24-bit RGB memory buffer without lock contention.

10.3.2 Tone-Mapping and Gamma 2.2 Correction

Physical ray tracing evaluates linear radiance. To map high-dynamic-range radiance onto standard display monitors, the CPU tracer clamps values and applies standard $\gamma = 2.2$ sRGB power-law encoding:

$$C_{\text{sRGB}} = \text{pow} \left(\text{clamp}(C_{\text{linear}}, 0.0, 1.0), \frac{1.0}{2.2} \right) \times 255.0 \quad (10.3)$$

The rendered image is saved to `raytraced_snapshot.bmp` via `TextureGenerator::writeBMP24`.

10.4 Performance Comparison

Table 10.1 compares the execution characteristics of the three rendering pipelines supported in *Matsuri Nights*.

Pipeline Architecture	Frame Rate	Shadow Fidelity	Reflection Capability
Standard Forward Rasterization (<code>basic.vert / basic.frag</code>)	60.0 FPS (16.6ms)	16-Sample PCF Soft Shadows (Depth FBO)	Local Blinn-Phong specular highlights; no inter-object reflection.
Real-Time GPU Ray Tracer (<code>raytrace.frag</code>)	32.5 FPS (30.7ms)	Analytic Hard Shadow Rays per light	Recursive 3-bounce Whitted mirror reflections on gold box, orb, and floor.
Multi-Threaded CPU Ray Tracer (<code>src/RayTracer.h</code>)	Offline (1.85s)	Analytic Hard Shadow Rays per light	Pristine, deterministic multi-bounce mirror reflections exported to BMP.

Table 10.1: Performance and Feature Comparison of the Three Rendering Pipelines.

11. User Interface, Heads-Up Display, and Context Interaction

11.1 In-Window HUD Architecture

Early graphics prototypes frequently clutter the developer console with stdout logs, disrupting visual immersion. To provide professional, in-engine telemetry with negligible rendering overhead, *Matsuri Nights* features an **In-Window Minimal Heads-Up Display (HUD)** implemented in `src/ui/Hud.cpp`.

11.1.1 Embedded 256 × 256 Consolas Font Atlas

To avoid complex external TrueType font parsing libraries (such as FreeType), the engine embeds an auto-generated 256 × 256 monochrome Consolas Bold font atlas as a static C++ array (`src/ui/FontAtlasData.h`):

- **Dimensions:** 256 width, 256 height (65,536 bytes).
- **Glyph Grid:** 16 columns × 6 rows covering printable ASCII characters (32...127).
- **Cell Dimensions:** Cell width 16 pixels, cell height 24 pixels.
- **OpenGL Texture:** Uploaded once as a single-channel GL_RED luminance texture.

11.1.2 Batched 2D Orthographic Render Pass

The HUD pass renders directly into the default framebuffer immediately prior to buffer swapping:

1. Depth testing is temporarily disabled (`glDisable(GL_DEPTH_TEST)`), and alpha blending is enabled (`glBlendFunc(GL_SRC_ALPHA, GL_ONE_MINUS_SRC_ALPHA)`).
2. The 2D vertex shader (`shaders/hud.vert`) applies an orthographic matrix mapping pixel coordinates $(x, y) \in [0, W] \times [0, H]$ onto NDC $[-1, 1] \times [1, -1]$:

$$x_{\text{ndc}} = \frac{2x}{W} - 1.0, \quad y_{\text{ndc}} = 1.0 - \frac{2y}{H} \quad (11.1)$$

3. **Batched VBO:** Both translucent dark background panels and textured glyph quads are packed into a single interleaved vertex buffer:

```
struct HudVertex {  
    float x, y;           // Screen pixel position  
    float u, v;           // Atlas UV (or -1.0 for solid colored panels)  
    float r, g, b, a;     // RGBA color tint & opacity  
};
```

A single `glDrawArrays` call dispatches the entire UI overlay in less than 0.05 ms.



Figure 11.1: In-window minimal HUD overlay displaying real-time FPS telemetry, active scene states, selected target metadata, and contextual control hints.

11.2 Context-Sensitive Interaction System

The interaction manager (`src/ui/InteractionManager.cpp`) tracks player position and spatial orientation relative to scene entities:

- **Proximity Auto-Selection:** When the player walks within interaction range of any building, stall, or landmark, the entity is automatically highlighted on the HUD.
- **Contextual Action Hints:** When standing near a townhouse, the HUD dynamically displays:

[H] Slide Open/Close Shoji Door [G] Slide Open/Close Windows

When approaching the stage, it displays [M] Replay Magic Show Trick.

- **Manual Selection Lock (Key T):** Pressing **T** locks selection onto a specific target, enabling live transformation manipulation. Pressing **Shift + T** unlocks selection back to proximity auto-mode.

11.3 Live In-Engine Object Transformation Controls

When an object is selected (via Key **T** or proximity), the user can translate, rotate, and scale it directly in the running viewport:

- **3D Translation:** Keys **I / K** modify Elevation ($\pm Y$), **J / L** modify Lateral Axis ($\pm X$), and **U / O** modify Depth ($\pm Z$).
- **3D Rotation:** **Arrow Keys Up / Down** rotate Pitch ($\pm X$); **Arrow Keys Left / Right** rotate Yaw ($\pm Y$).
- **3D Scale:** Keys **+ / -** (or **[/]**) scale the object by $\pm 10\%$.

All modifications update local transform matrices immediately and propagate across child nodes, demonstrating real-time scene graph reactivity.

12. Verification, Testing, and Automated Quality Assurance

12.1 The Need for Automated Testing in Graphics

Graphics applications are notoriously difficult to test reliably. Visual inspection alone easily misses subtle matrix regressions, bounding box leaks, broken uniform state caching, or inverted trigonometric signs. To guarantee flawless stability and continuous quality assurance, *Matsuri Nights* features an extensive **Automated Test Suite** executed via the `--test` command-line flag.

12.2 The 312-Test Automated Test Harness

When invoked with `--test`, the application initializes a hidden OpenGL context (`glfwWindowHint(GLFW_VISIBLE, GLFW_FALSE)`) and executes `runAutomatedTestSuite()` (`Main.cpp`: lines 510--960). The test suite verifies 312 discrete unit assertions partitioned across five specialized domains:

Test Domain	Test Count	Verified Architectural Behaviors
Section 1: Inspectable Objects & Transforms	106 Tests	Iterates through all 15 registered inspectable objects. For every object, verifies node validity, +X translation, return to origin, +Y elevation, return, +Z depth, return, Pitch rotation, return, Yaw rotation, return, and scale up/down.
Section 2: Interactive Doors & Windows	40 Tests	Verifies that all 4 Machiya townhouses (L1, L2, R1, R2) start closed, slide open on Key H, progress incrementally, slide closed, disallow interaction beyond 5 meters, and operate sliding Shoji windows on Key G.
Section 3: Environment, Shading & Physics	36 Tests	Tests Day/Night state toggle, animation pause and resume, shading mode cycle (0 → 1 → 2 → 0), texture map toggle, shadow mapping toggle, GPU ray tracing toggle, wall collision blocking, and camera presets (1, 2, 3, Reset R).
Section 4: Festival Stall Animations	12 Tests	Asserts Takoyaki stall existence, ball count (== 6), hopping trajectory progression, Kakigori machine existence, bowl count (== 6), and ice shaver flywheel rotation.
Section 5: In-Window HUD & Viewport Export	28 Tests	Verifies HUD visibility toggle (F1), auto-selection proximity tracker, manual selection lock (Key T), unlock to auto (Shift+T), lantern light toggle (Key O), magic trick replay (Key M), and exports of both 24-bit BMP and high-resolution PNG viewport screenshots.
Total Test Assertions	312 Tests	312 / 312 Tests Passed (100% Success Rate)

Table 12.1: Structure of the 312-Test Automated Verification Suite (`Main.cpp`).

12.2.1 Console Output of the Test Runner

```
=====
MATSURI NIGHTS - AUTOMATED COMPREHENSIVE CONTROL & TRANSFORM TEST SUITE
Verifying All Controllable Objects, Interactive Elements & Toggles
=====

--- [SECTION 1: CONTROLLABLE INSPECTABLE OBJECTS & TRANSFORMS] ---
[PASS] Inspectables List Count == 15 (Count = 15)
... [15 objects tested x 7 transform assertions each = 105 tests pass] ...

--- [SECTION 2: INTERACTIVE HOUSE ENTRANCE DOORS & SHOJI WINDOWS] ---
... [4 houses tested for door/window toggles, sliding, distance = 40 tests pass] ...

--- [SECTION 3: ENVIRONMENT, RENDERING & LIGHTING TOGGLES] ---
... [Day/Night, Pause, Shading 0->1->2->0, Shadows, Ray Tracing, Collision = 36 tests pass] ...

--- [SECTION 4: FESTIVAL STALL ANIMATIONS] ---
... [Takoyaki 6 balls, Kakigori 6 bowls, flywheel, ice block = 12 tests pass] ...

--- [SECTION 5: IN-WINDOW HUD OVERLAY & CONTEXT INTERACTION SYSTEM] ---
... [HUD update, lock/unlock selection, BMP & PNG exports = 28 tests pass] ...

=====
TEST SUMMARY: 312 / 312 TESTS PASSED (100% SUCCESS)
=====
```

12.3 Automated Report Figure Capture Harness

To eliminate manual screenshot taking and ensure that every screenshot embedded in this academic report matches the exact programmatic state of the code, *Matsuri Nights* implements the `--capture-report` CLI command:

1. Spawns a hidden OpenGL context with resolution 1280×720 .
2. Iterates sequentially through twenty predefined photographic shot configurations (`ss_01` through `ss_20`).
3. For each shot, automatically sets exact camera position, yaw, pitch, zoom, day/night lighting factor, shading mode, shadow state, texture state, and animation hooks.
4. Renders the scene (and optional HUD overlay), finishes draw calls, and captures viewport pixels via `glReadPixels`, encoding them directly to PNG using `stbi_write_png`.
5. Saves the images to `report/figures/`.

This ensures 100% mathematical reproducibility: any examiner or developer can re-generate the figures in seconds by running `Matsuri Nights.exe --capture-report`.

13. Controls Reference and Documentation Audit

13.1 Complete Interactive Controls Reference

Table 13.1 provides the authoritative reference for all user inputs, keyboard hotkeys, and mouse interactions supported by *Matsuri Nights*.

Category	Key / Input Binding	Action / Engine Functionality
Camera Movement	W / A / S / D	Move camera Forward, Left, Backward, Right
	E / Q	Elevate camera Up / Descend Down
	Left Shift	Sprint modifier ($2.5\times$ velocity boost)
	Mouse Drag	First-person 6-DOF look around (Pitch & Yaw)
	Mouse Wheel	Adjust camera field-of-view (Zoom $1^\circ \dots 60^\circ$)
	C	Toggle mouse cursor capture (free mouse cursor)
Camera Presets	R	Reset camera position and orientation to default origin
	1	Preset 1: Festival street entrance (0.0, 3.5, 26.0)
	2	Preset 2: Magic stage & audience close-up (6.2, 2.2, -10.5)
	3	Preset 3: Grand Torii gate and sky view (0.0, 2.5, -18.0)
Environment & Shading	N	Toggle Day / Night celestial cycle ($0.0 \leftrightarrow 1.0$)
	Spacebar	Pause / Resume all scene kinematic animations
	0 / Numpad 0	Toggle lantern and stall illuminations (ON / Dimmed)
	P	Cycle Shading Mode (Blinn-Phong $\rightarrow$ Diffuse $\rightarrow$ Ambient)
	V	Toggle 16-sample PCF soft shadow mapping (ON / OFF)
	X	Toggle diffuse texture mapping (ON / Material Colors)
	Z	Toggle real-time GPU Whitted ray-tracing mode (ON / OFF)
	F9	Trigger multi-threaded CPU ray-traced snapshot to BMP
Context & Interaction	F	Launch manual fireworks rocket into nocturnal sky
	F1	Toggle in-window minimal HUD overlay (Show / Hide)
	F10	Capture clean high-resolution viewport screenshot
	Shift + F10	Capture high-resolution viewport screenshot with HUD
	T	Cycle next target object & lock manual transform mode
	Shift + T	Unlock manual target $\rightarrow$ return to proximity auto-mode
	H	Slide open / close nearest house front Shoji door
	G	Slide open / close nearest house sliding Shoji windows
Live Object Transforms	B	Toggle Wall Collision (Walk Mode $\leftrightarrow$ Noclip Flight)
	M	Replay magic show trick sequence from Phase 1
	I / K, J / L, U / O	Translate selected object ($\pm Y, \pm X, \pm Z$)
	Arrow Keys	Rotate selected object (Up/Down: Pitch, Left/Right: Yaw)
	+ / - or [/]	Scale selected object ($\pm 10\%$ uniform scale)

Table 13.1: Authoritative Keybinding and Interaction Catalog (Main.cpp: `key_callback()`).

13.2 Documentation Discrepancies and Audit

During the thorough audit of the project repository, discrepancies were identified between legacy markdown documentation files (Plan.md, Details.md, controls.md, color_changes.md, and README.md) and the final ground-truth source code. Table 13.2 provides the authoritative reconciliation.

Feature / Metric	Legacy Doc Statement	Ground-Truth Source Code	Code Citation
Point Light Count	Stated as 12 point lights (or 14 total lights).	14 dynamic point lights (0–13) plus 1 Directional and 1 Spotlight = 16 total lights .	shaders/basic.frag: line 44 src/Scene.h: line 646
Inspectables Count	Stated as 10 or 11 inspectable objects.	15 inspectable objects registered in inspectables array (indices 0–14).	src/Scene.h: lines 880--898 Scene::setupInspectables()
Screenshot Hotkey	Early planning suggested Key P for screenshots.	Key P is bound to Cycle Shading Mode . Screenshots are triggered by F10 and Shift+F10 .	Main.cpp: lines 324--331 Main.cpp: lines 380--381
Light Toggle Key	Early notes used Key L to toggle lights.	Remapped to Key 0 and Numpad 0 to preserve J/L for $\pm X$ translation.	Main.cpp: lines 357--361
C++ Standard Dialect	Some markdown sections referenced C++17.	The project explicitly compiles under ISO C++20 (/std:c++20).	Matsuri Nights.vcxproj: line 77
Door Interaction Range	Early notes implied global interaction.	Interaction is restricted to player proximity within 5.0 meters .	src/Scene.h: lines 1525, 1570
CPU Ray Tracer Export	Legacy notes suggested PNG or PPM.	Native CPU tracer writes uncompressed 24-bit BMP (writeBMP24).	src/RayTracer.h: line 655

Table 13.2: Reconciliation of Legacy Documentation Discrepancies Against Code Truth.

14. Conclusion and Future Enhancements

14.1 Summary of Technical Achievements

The development of “*Matsuri Nights — A Japanese Festival Street*” represents a comprehensive exploration of modern real-time computer graphics. Built completely from the ground up using ISO C++20 and Modern OpenGL 3.3 Core Profile, the project fulfills and surpasses all academic criteria mandated by the CSE4102 curriculum at KUET:

1. **Mathematical Rigor:** Exact implementation of hierarchical TRS transformations, normal matrix inversion, Gram-Schmidt camera orthonormalization, and Bishop frame parallel transport.
2. **Complex Moving Hierarchies:** Seamless kinematic motion where child nodes (lantern bodies, magic orbs, pedestrian limbs) evaluate physics relative to moving parent coordinate systems.
3. **Advanced Optical Illumination:** A forward rendering pipeline managing 16 active dynamic lights with distance attenuation, spotlight cutoff cones, dynamic sky dome transitions, and 16-sample PCF soft shadow mapping.
4. **Dual Ray Tracing Extensions:** Integration of both an interactive real-time GPU Whitted ray tracer operating on full-screen quads and a multi-threaded CPU snapshot ray tracer.
5. **Automated Verification:** 312 unit test assertions passing with 100% success and an automated report figure capture harness ensuring complete reproducibility.

14.2 Future Enhancements

While *Matsuri Nights* delivers exceptional visual fidelity and real-time responsiveness ($> 60\text{FPS}$), several architectural extensions present exciting avenues for future research:

- **Physically-Based Rendering (PBR):** Migrating from Blinn-Phong empirical reflectance to the microfacet Cook-Torrance BRDF with the GGX normal distribution function, metallic-roughness parameterization, and Image-Based Lighting (IBL) via cubemaps.
- **Normal and Parallax Occlusion Mapping:** Utilizing tangent space normal maps and height fields to simulate 3D relief displacement across stone cobblestones and ceramic roof tiles without increasing polygonal geometric complexity.
- **Volumetric Fog and Light Shafts (God Rays):** Ray-marching participating media through the 2048×2048 shadow depth map to render atmospheric haze and volumetric light shafts emanating from lanterns.

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